

LAB MANUAL : PROTEIN BIOMANUFACTURING AND BIOCHEMISTRY (BIOT274 IVC)



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TABLE OF CONTENTS

Licensing

- 1: Activity 1-1 - Protein Extraction Using Bacterial Cell Lysis
- 2: Activity 1-2 - Bradford Assay – Measuring Protein Concentration
- 3: Activity 1-3 - Beer's Law Spectrophotometric Analysis
- 4: Activity 1-4 - Analysis of Enzymatic Activities of Cell-Free Extracts
- 5: Activity 2-1 - Purification via Ammonium Sulfate Saturation of Cell-Free Extract
- 6: Activity 2-2 - Dialysis for Enzyme Purification
- 7: Activity 2-3 - Size Exclusion Gel Filtration Purification of Enzymes
- 8: Activity 2-4 - Verification of Protein Purification via Polyacrylamide Gel Electrophoresis
- 9: Activity 3-0 - Project Instructions - Investigating Enzyme Kinetics
- 10: Activity 3-1 - Michaelis-Menten Analysis of Substrate Concentration on Reaction Rate
- 11: Activity 3-2 - Determining the IC_{50} of Inhibitor
- 12: Activity 3-3 - Identifying the Type of Enzyme Inhibition

Materials

- [Index](#)
- [Glossary](#)
- [Detailed Licensing](#)
- [Detailed Licensing](#)

Licensing

A detailed breakdown of this resource's licensing can be found in [Back Matter/Detailed Licensing](#).

1: Activity 1-1 - Protein Extraction Using Bacterial Cell Lysis

Learning Objectives

- Explain how lysozyme and freeze-thaw cycles contribute to bacterial cell lysis.
- Describe the function of each reagent and step in the lysis protocol.
- Understand the rationale behind using controlled temperature conditions.
- Predict potential outcomes of improper lysis (e.g., protein degradation, incomplete lysis).
- Prepare and label bacterial cultures for protein extraction properly.

Definition: Term

- **Cell Lysis:** The breaking down of a cell's membrane to release intracellular contents.
- **Lysozyme:** An enzyme that hydrolyzes β -1,4-glycosidic bonds in peptidoglycan, weakening bacterial cell walls.
- **Peptidoglycan:** A mesh-like polymer of sugars and amino acids that forms the bacterial cell wall.
- **Freeze-Thaw Cycle:** A physical method of lysis where repeated freezing and thawing forms ice crystals that disrupt membranes.
- **Non-ionic Detergent:** Mild surfactants used to solubilize membranes without denaturing proteins.
- **Cell-Free Extract (CFE):** The supernatant containing soluble proteins after lysis and centrifugation.

Note

- Review bacterial cell structure, especially the cell wall and membrane.
- Bring your lab notebook with a flowchart.
- Be ready to work with cold reagents and keep enzymes on ice at all times.

Project 1a - Protein Extraction Using Bacterial Cell Lysis

Cell lysis via Lysozyme and Freeze-Thaw Cycle

Background:

The Bacterial Cell Lysis Kit is designed for the efficient extraction of soluble proteins from bacterial cells using an enzyme-assisted chemical lysis approach. The primary method of cell disruption relies on **lysozyme digestion** of the bacterial cell wall. **Lysozyme** specifically cleaves the β -1,4-glycosidic linkages in the peptidoglycan cell wall, thereby weakening the integrity of the bacterial wall, particularly in Gram-negative species like *E. coli*. This enzymatic step is enhanced with **non-ionic detergent**, which is also part of the buffer, which helps disrupt the cytoplasmic membrane. The buffer also contains a mixture of **organic buffers and salts** that stabilize proteins and reduce aggregation or denaturation of proteins during extraction.

Before applying the lysis buffer, you will subject the bacterial pellet to **three freeze-thaw cycles**. This step involves repeatedly freezing the cell suspension (typically at -20°C or -80°C) and then thawing it at room temperature or 37°C . Freeze-thawing helps initiate cell lysis by forming ice crystals that puncture cellular membranes and disrupt internal structures. Repeating this process three times significantly weakens the cell envelope, making the cells more susceptible to enzymatic and detergent-mediated lysis. It also facilitates better access of lysozyme and detergent to the peptidoglycan layer and inner membrane, improving the efficiency and completeness of protein extraction. Importantly, this method is gentle and preserves protein integrity for downstream functional or structural analysis.

After lysis, the mixture is clarified by **low-speed centrifugation**, allowing for the separation of the **soluble cell-free extract (CFE)** from insoluble components such as cell debris and inclusion bodies. The resulting supernatant is suitable for various downstream applications, including **affinity chromatography, SDS-PAGE, protein quantification, purification, and enzyme kinetic and drug studies**.

Preparation (Before Class)

1. Prepare four (4) **sterile 1L LB broths** with 100 $\mu\text{g/mL}$ **ampicillin** (LB + Amp).
2. Inoculate **three flasks** with **pT7BM3 *E. coli*** into three of the LB + Amp flasks.

3. Inoculate **one flask** with **pET3a *E. coli*** into the remaining LB + Amp flask.
 4. **Label each flask clearly** with the Name and date.
 5. Incubate all four flasks at **32°C with shaking** overnight.
 6. After ~24 hours of incubation, store the cultures at **4°C** in the Lab fridge until use (**Can be stored up to a week**)
-

Materials

- (From preparation step) *E. coli* cultures (BL21) with **pT7BM3 or pET3a plasmids**
- **Disinfectant spray, biohazard tip waste, Biohazard liquid waste container**
- **Centrifuge** with 50 mL rotor capacity
- **50 mL conical tubes** (Corning™352070 or similar)
- **-80°C freezer or dry ice**
- **37°C shaking incubator**
- **Vortex mixer**
- **Bacterial Cell Lysis Kit** (GoldBio #GB-176)
 - Lysis Buffer (stored at **4°C**)
 - Lysozyme (stored at **-20°C**)
- **Ice bucket** (keep filled with ice)

Important Reminder: Keep **all enzyme-containing tubes on ice** throughout the experiment unless directed otherwise. Cold temperatures help maintain enzyme activity and prevent denaturation. You'll be given an individual ice bucket to use at your bench.

Collecting Cell Pellet

1. Label and weigh six 50 mL conical tubes. Record the weight:

Note: The Instructor may change the protocol at their discretion.

- Tube 1: _____ g
 - Tube 2: _____ g
 - Tube 3: _____ g
 - Tube 4: _____ g
 - Tube 5: _____ g
 - Tube 6: _____ g
2. Pour the corresponding cultured bacteria into each labeled tube. Ensure that **final tube weights match** (± 0.5 g) by adjusting volumes so they're roughly equal (balance is important for centrifugation).
 3. Centrifuge tubes at **5000 rpm for 7 minutes** at 4°C.
 4. Carefully **discard** the supernatant without disturbing the pellet.
 5. Repeat steps 2–4 until there's no more bacteria remaining.

Freeze-Thaw Lysis (Pre-Lysis Step)

1. Place cell pellets in **dry ice or -80°C freezer** for ~5 minutes (until fully frozen). (If you have a -20°C freezer, then leave it in for 15 minutes)
2. Thaw samples in a **37°C shaking incubator** for ~7 minutes.
3. Repeat the freeze-thaw cycle **three times in total**.

Cell Lysis

1. Add **5x volume of lysis buffer** based on pellet weight
 - Example: your conical tube pre-weight was 50 g, and your new weight is 51 g, and thus, you have 1g of pellet. → Add 5 mL lysis buffer.
2. Add **10 μ L lysozyme for every 1 mL** of lysis buffer added.
3. Record the pellet weights, buffer volumes, and lysozymes added:
 - Pellet 1: _____ g → _____ mL buffer → _____ μ L lysozyme
 - Pellet 2: _____ g → _____ mL buffer → _____ μ L lysozyme

- o Pellet 3: _____ g → _____ mL buffer → _____ μ L lysozyme
- o Pellet 4: _____ g → _____ mL buffer → _____ μ L lysozyme
- o Pellet 5: _____ g → _____ mL buffer → _____ μ L lysozyme
- o Pellet 6: _____ g → _____ mL buffer → _____ μ L lysozyme

4. Use a p1000 pipette to mix up-and-down **gently** to resuspend each pellet. **⚠ Avoid bubbles!** Use fresh tips if clogged and pipette slowly.
5. Place tubes in a **37°C shaking incubator** for **1 hour** to allow enzymatic lysis.
6. Vortex each lysate for **30 seconds** to help homogenize the lysed solution.

Collect the Soluble Extract (CFE)

1. Centrifuge tubes at **5000 rpm for 30 minutes** at 4°C. Make sure tubes are balanced in the centrifuge!
2. Carefully collect the **supernatant** (the liquid portions that contain your soluble proteins).
3. Combine all the CFE-B supernatants, and then aliquot into fresh labeled tubes:
 1. 1 mL → Label as **"Project 1"**
 2. Split the remaining supernatant into two parts, and label one **"Project 2"** and the other **"Project 3"**
4. Record and compare the **color or appearance** of each sample.

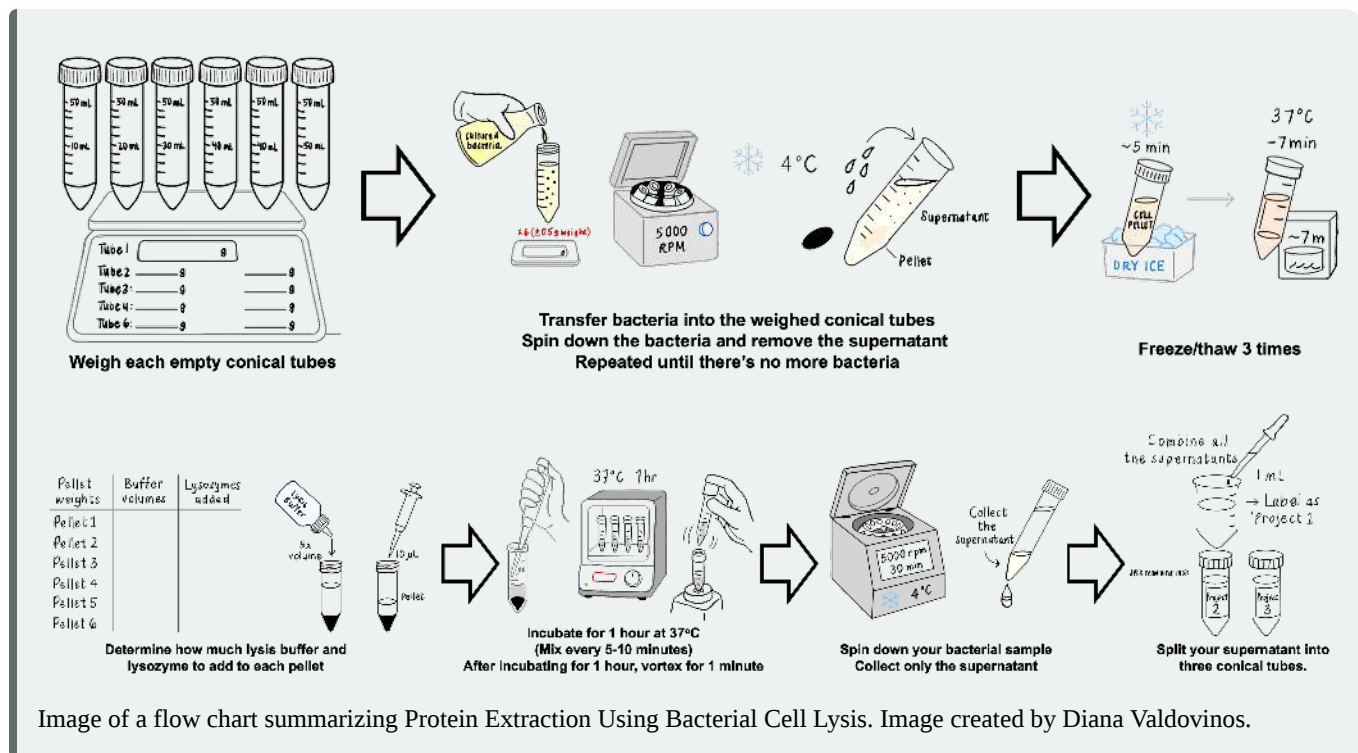


Image of a flow chart summarizing Protein Extraction Using Bacterial Cell Lysis. Image created by Diana Valdovinos.

Learning Objectives

- Compare physical and enzymatic methods of cell lysis.
- Justify why lysozyme is effective in lysing Gram-negative bacteria like *E. coli*.
- Analyze how different steps (e.g., temperature, timing, mixing) impact protein extraction success.
- Reflect on best practices in sterile technique and reagent handling.

Reflection Questions

1. How did the freeze-thaw process aid the lysis? What physical changes did you observe?
2. Why is lysozyme particularly effective against *E. coli*? Would this work the same for Gram-positive bacteria?
3. What steps in the protocol are most sensitive to timing or temperature? Why?
4. Did your samples differ in color or appearance? What could this indicate about protein content or contamination?

5. How might incomplete lysis affect downstream protein quantification or purification?

 Note

1. Complete and submit your lab worksheet with recorded weights, volumes, lysozyme added, and CFE descriptions.
2. Include brief written answers to reflection questions.
3. Label and submit clearly marked CFE tubes ("Bradford and Beer", "CFE-A/B Part 1/2").

1: [Activity 1-1 - Protein Extraction Using Bacterial Cell Lysis](#) is shared under a [not declared](#) license and was authored, remixed, and/or curated by LibreTexts.

2: Activity 1-2 - Bradford Assay – Measuring Protein Concentration

Learning Objectives

- Explain the principle of the Bradford protein assay and how it detects protein concentrations.
- Understand why measuring protein concentration is essential before assessing enzyme activity.
- Describe the structure and function of cytochrome P450 enzymes, particularly P450_{BM3}.
- Identify the significance of using a standard curve in quantitative spectrophotometry.
- Prepare reagents and samples accurately using serial dilution techniques.

Definition: Term

- **Bradford Assay:** A colorimetric assay that uses Coomassie Brilliant Blue G-250 to detect and quantify protein concentrations.
- **Coomassie Brilliant Blue:** A dye that binds to proteins, especially basic and aromatic residues, causing a shift in absorbance from 465 nm to 562 nm.
- **Standard Curve:** A graph of known concentrations used to determine the concentration of unknown samples.
- **Cytochrome P450 (P450s):** A family of heme-containing enzymes that catalyze oxidation of substrates including drugs and toxins.
- **P450_{BM3}:** A self-sufficient bacterial cytochrome P450 enzyme from *Bacillus megaterium*, containing fused reductase and heme domains.
- **Absorbance (A):** A measure of the amount of light absorbed by a solution at a specific wavelength.
- **Cell-Free Extract (CFE):** A lysate containing soluble cellular components, including proteins, from which intact cells have been removed.
- **Dilution Factor:** The ratio by which a solution is diluted; in this lab, samples are diluted 10-fold.

Note

- Prepare a dilution series of BSA to generate a standard curve.
- Perform the Bradford assay on CFE-A and CFE-B.
- Use a spectrophotometer to measure absorbance at 562 nm.
- Plot a standard curve and determine unknown protein concentrations.

Week 1b: Bradford Assay – Measuring Protein Concentration

Cytochrome P450 enzymes are a large and diverse family of heme-containing monooxygenases that play critical roles in the **metabolism of drugs, toxins**, and endogenous compounds across all domains of life. In humans, these enzymes are involved in **detoxification** and **drug metabolism in the liver**, while in microorganisms, they often help degrade environmental chemicals or participate in specialized metabolic pathways. Because of their catalytic versatility, cytochrome P450s are widely studied in biotechnology for applications such as **drug development, bioremediation**, and **synthetic biology**. In this lab, we focus on **P450_{BM3}** from *Bacillus megaterium*, a well-characterized bacterial P450 that is especially useful for educational and research purposes. Unlike most P450s that require separate reductase proteins, P450_{BM3} is a naturally fused, self-sufficient enzyme consisting of a 119 kDa polypeptide with FAD, FMN, and heme cofactors all in one protein. This fusion allows for efficient electron transfer from NADPH to FAD, then to FMN, and finally to the heme center, where oxygen activation and substrate oxidation occur. The heme group binds oxygen and substrate, producing a characteristic absorption peak at 418 nm, a spectral signature used to detect the presence of active P450_{BM3}.

For this week's experiments, two cell-free extracts have been prepared from *E. coli*: one expressing an empty vector plasmid (pET3) and the other expressing the P450_{BM3} gene (pT7_{BM3}). Only the extract containing P450_{BM3} will show characteristic P450 activity. To investigate this, students will perform three spectrophotometric assays: (1) the **Bradford assay** to estimate total protein concentration using Coomassie Brilliant Blue G-250 dye, which shifts absorbance from 465 nm to 562 nm when bound to proteins—mainly through interactions with basic and aromatic residues; (2) an **absorption scan** to detect the 418 nm peak associated with

the heme group of cytochrome P450_{BM3}; and (3) an **enzyme activity assay** to monitor NADPH oxidation by measuring the decrease in absorbance at 340 nm.

The **Bradford assay** will provide an **estimate of protein concentration in the extracts**, although it does not distinguish between P450_{BM3} and other cellular proteins. The combined results will help students visualize and quantify the presence and activity of a bacterial cytochrome P450 enzyme, laying the foundation for understanding how these enzymes function and why they are important in both natural and applied contexts.

In this lab, you will use the Bradford assay to estimate protein concentrations in two cell-free extracts (CFE-A and CFE-B) using a standard curve generated from known concentrations of bovine serum albumin (BSA). This is an essential first step in protein studies and helps us normalize enzyme activity in later experiments.

Materials

- **Pierce BCA Protein Assay Kit** (Thermo Scientific, #23227)
 - (Contains BSA standard [2 µg/µL], Reagent A, and Reagent B)
- Plastic cuvettes (at least 10)
- Distilled water
- **Cell-free extracts** A and B (from previous lab)
- 15-mL or 50-mL conical tubes
- 37°C incubator
- Spectrophotometer
- Pipettes and tips
- Parafilm (to seal cuvettes)
- Scissors, Sharpies, acetone, cotton swabs

Protocol

1. Thaw your **CFE** from the freezer (This is your protein extract after bacterial lysis).
2. **IMPORTANT: Calculate** how much **Reagent A** and **Reagent B** to mix.
 - You will need 500 µL of Reagent A per sample.
 - You will need 5 µL of Reagent B per sample
 - For example, if you have 10 samples, then you will mix 5 mL of Reagent A with 50 mL of Reagent B.
3. **IMPORTANT:** In a 15-mL conical tube, mix your calculated **Reagent A** with **Reagent B** (**50:1** ratio).
4. **IMPORTANT:** Label this tube **"Bradford Reagent"**.
5. Obtain your cuvettes* (This may change based on your instructor's protocol)
 - **1 cuvettes:** "CFE"
 - **5 cuvettes:** "BSA standard" (#1 to #5)
 - **1 cuvettes:** "Blank" (contains no BSA)
 - Add 50 µL of buffer to this cuvette.
6. Use the **serial dilution method** to generate known concentrations:
 - Label the "BSA standard" cuvettes #1 through #5.
 - Add 100 µL of BSA standard (2 µg/µL) to **cuvette #1**.
 - Add **50 µL of water** to cuvettes #2 to #5.
 - Then take 50 µL from #2 to #3, mix.
 - Continue: #3 to #4, then #4 to #5.
 - Discard 50 µL from cuvette #5 after mixing.
 - Transfer **50 µL** from cuvette #1 to #2, mix well.
7. Mix **5 µL of your CFE sample** + **45 µL of buffer** (this is a 10-fold dilution)
8. To all your cuvettes (standards, samples, and blank), add **450 µL of Bradford reagent**
9. Place all cuvettes in a 37°C incubator, and incubate for **5 minutes** (you can go longer for deeper color development).
10. **Blank with cuvette your "Blank"**
11. Measure the absorbance of each standard and sample.
12. Rinse cuvettes **first with ethanol**, then with **distilled water**, and allow them to air dry.

Concentration (mg/mL)	Absorbance
2	
1	
0.5	
0.25	
0.125	
Sample CFE	

Create your standard curve

- On Excel or Sheets, plot:
 - X-axis:** BSA concentration ($\mu\text{g}/\mu\text{L}$)
 - Y-axis:** Absorbance at 562 nm
- Add a **trendline** and **display the equation and R^2 value**.
- Use the equation to calculate the protein concentration in **CFE-A and CFE-B**, remembering to account for the 10-fold dilution.

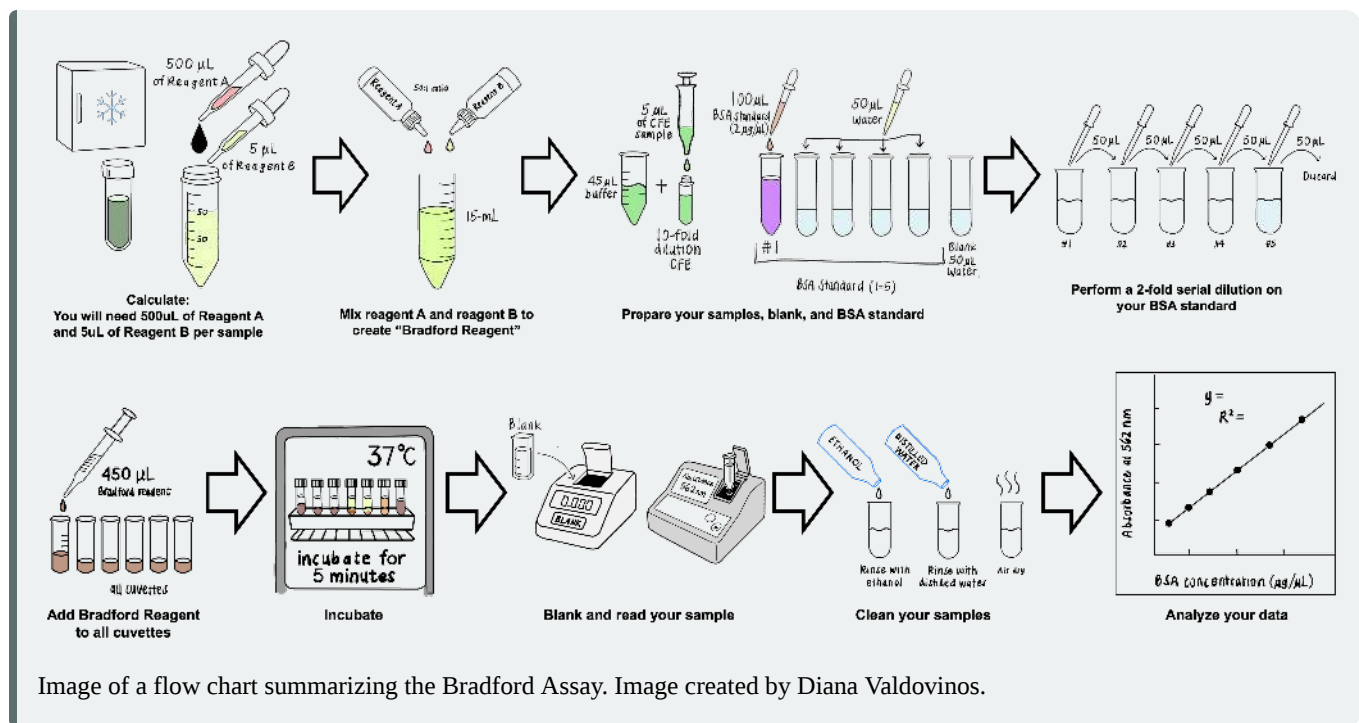


Image of a flow chart summarizing the Bradford Assay. Image created by Diana Valdovinos.

Reference

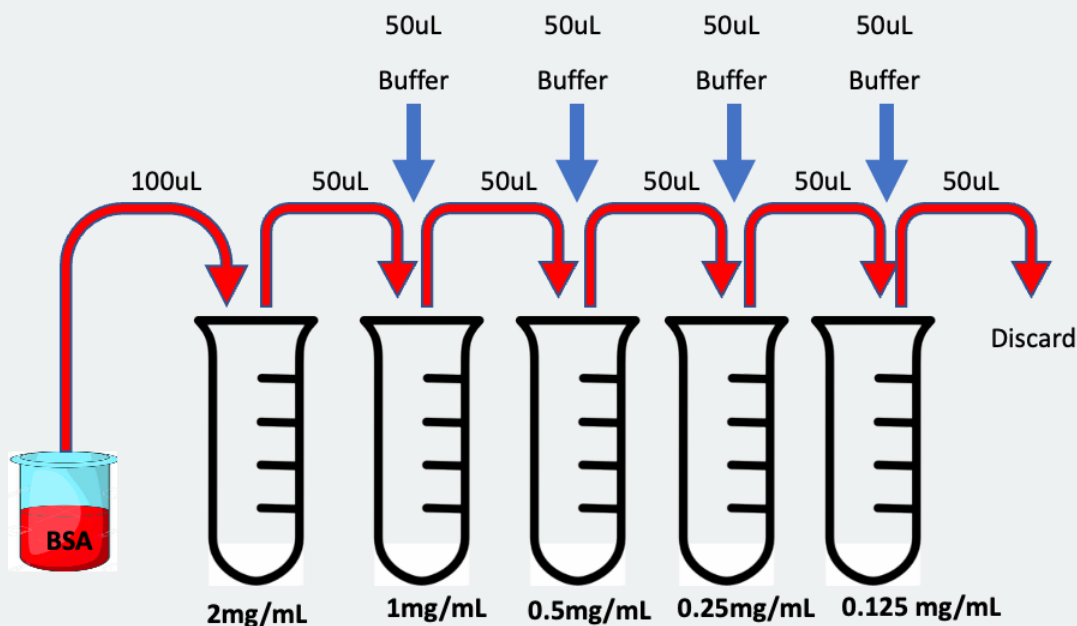


Image of 2-fold serial dilution created by Victor Pham.

Learning Objectives

- Accurately interpret a standard curve to calculate protein concentrations.
- Evaluate the reliability of spectrophotometric data using R^2 values from trendlines.
- Reflect on the limitations of the Bradford assay (e.g., protein-to-protein variability).
- Discuss why accurate protein quantification is foundational for normalizing enzymatic activity.
- Connect the Bradford assay to future assays detecting cytochrome P450 function (absorption at 418 nm and NADPH consumption at 340 nm).

Key Reflections

Prompt	Student Response Example
What did the standard curve tell you about your samples?	"It helped us estimate the protein concentration of CFE-A and CFE-B based on their absorbance."
How reliable was your data (based on R^2 value)?	"Our R^2 value was 0.99, suggesting a strong linear relationship between BSA concentration and absorbance."
Did both CFEs show similar protein concentrations? Why might they differ?	"They were different, possibly because only one sample expresses the P450 _{BM3} protein."
What limitations might affect the Bradford assay's accuracy?	"It measures total protein, so it doesn't tell us how much of that protein is P450 _{BM3} ."
How will knowing protein concentration help in later P450 assays?	"We need to normalize enzyme activity to protein amount to compare true enzyme efficiency."

Post-Lecture Questions

1. What wavelength is used to measure the Bradford assay and why?
2. How would you troubleshoot if your blank cuvette had an unexpectedly high absorbance?
3. If two extracts have different protein concentrations, how might that affect their enzyme activity measurements?
4. What does an R^2 value of 0.85 tell you about your standard curve?
5. In the context of drug metabolism, why is it important to measure the expression and activity of enzymes like cytochrome P450?

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3: Activity 1-3 - Beer's Law Spectrophotometric Analysis

Learning Objectives

- Explain the principle of Beer's Law and how it relates absorbance to concentration.
- Identify the components and functions of a spectrophotometer.
- Describe the structural and functional significance of the P450_{BM3} enzyme.
- Understand how light absorption by heme can be used as a proxy to determine enzyme concentration.

Definition: Term

- **Beer's Law (Beer-Lambert Law):** A relationship describing how the absorbance of light (A) is proportional to the concentration (c) of a substance, path length (L), and its extinction coefficient (ϵ): $A = \epsilon cL$.
- **Absorbance (A):** A unitless measurement of how much light is absorbed by a sample. Also referred to as optical density (O.D.).
- **Extinction Coefficient (ϵ):** A constant that reflects how strongly a substance absorbs light at a particular wavelength. Units: $M^{-1}cm^{-1}$.
- **Path Length (L):** The distance light travels through the sample in the cuvette. Usually 1 cm.
- **Cytochrome P450:** A large family of enzymes involved in oxidation reactions, many of which metabolize drugs or steroids. They contain a heme prosthetic group.
- **P450_{BM3}:** A bacterial cytochrome P450 enzyme from *Bacillus megaterium* that is self-sufficient and highly active, commonly used in biochemistry studies.
- **Heme:** An iron-containing porphyrin ring that binds oxygen and gives P450 enzymes their characteristic absorbance near 418 nm.
- **UV/Vis Spectrophotometer:** An instrument that measures how much UV or visible light a sample absorbs. Used to quantify analyte concentration.

Beer's Law Spectrophotometric Analysis of P450 Enzyme Expression

In biochemistry, the **spectrophotometer** is a critical tool used to analyze the concentration of molecules in solution, particularly those that absorb light at specific wavelengths. This analytical method is based on the **Beer-Lambert Law** (commonly referred to as Beer's Law), which describes the relationship between the absorbance of light by a substance and its concentration in solution. The fundamental equation for Beer's Law is expressed as:

$A = \epsilon cL$, where A is the **absorbance** (also known as optical density or O.D.), ϵ (epsilon) is the **molar extinction coefficient** (in units of $M^{-1}cm^{-1}$), c is the **concentration** of the absorbing species in moles per liter (M), and L is the path **length** of the light through the sample (usually in centimeters).

This law indicates that absorbance is directly proportional to both the concentration of the substance and the distance the light travels through the solution. For example, an absorbance of 1.0 corresponds to a tenfold decrease in transmitted light intensity compared to the incident light, meaning only 10% of the original light passed through the sample. The spectrophotometer measures the intensity of light before and after it passes through the solution, allowing us to determine the absorbance and, using Beer's Law, calculate the concentration of the molecule of interest.

In this lab, you will apply Beer's Law to measure the activity of a cytochrome P450 enzyme, specifically P450_{BM3}, which has been engineered and expressed in *E. coli* BL21 cells. Cytochrome P450 enzymes are heme-containing monooxygenases that play a central role in the metabolism of drugs, xenobiotics, and endogenous substrates. P450_{BM3}, derived from *Bacillus megaterium*, is a well-studied, self-sufficient P450 with high catalytic activity, making it an excellent model for enzyme kinetics and biocatalysis studies.

Objective: To measure the absorbance of cell-free extracts containing P450_{BM3} using UV-Vis spectrophotometry and to determine the concentration of the heme-bound P450 enzyme using Beer's Law.

Materials:

- **UVette Cuvettes** (Eppendorf, Cat. #952010051)
- **Control CFE** (without Heme or LB Broth only)
- **Cell-Free Extracts (CFE)**
- **Pipettes and Tips**
- **UV/Vis Spectrophotometer** (Thermo Scientific, Cat. #840-300000)
- **Extraction Buffer (50 mM potassium phosphate, pH 7.4) OR 0.25X diluted PBS**
 - Prepared by diluting from a 1 M stock solution:
 - 95 g monobasic potassium phosphate (Carolina, #884250)
 - 52.5 g dibasic potassium phosphate (Carolina, #884300)

Procedure:

1. Thaw your **CFE**.
2. Pipette **300 μ L** of **Control CFE** (LB Broth only) into a UVette cuvette. This will be your blank for spectrophotometer calibration.
3. Pipette **300 μ L** of **CFE** into another new UVette cuvette.
4. **Measure Absorbance:**
 - Set the spectrophotometer to **418 nm**, the wavelength where heme (bound to P450) absorbs light.
 - Use the **Control CFE cuvette** to **blank** the machine.
 - Then measure the absorbance of your **CFE**, and record your results.
5. After measurements, return any remaining sample from the cuvettes back to their original labeled tubes.
6. Freeze the tubes if you are not proceeding directly to the enzymatic assay.
7. Rinse cuvettes **first with ethanol**, then with **distilled water**, and allow them to air dry.

Analyze Bradford vs Beer's Law:

Beer's Law Equation:

$$A = \epsilon \cdot c \cdot L$$

- A = absorbance (from the spectrophotometer)
- ϵ = **extinction coefficient** = **91,000 $M^{-1}cm^{-1}$** (for heme at 418 nm)
- c = concentration (Molarity, M)
- L = **path length** = **1 cm** (standard cuvette)

Rearranging to solve for concentration: $c = \frac{A}{\epsilon \cdot L}$

Step-by-Step Math Example:

Suppose your absorbance reading at 418 nm is **0.11**.

1. Calculate concentration (M):

$$c = \frac{0.11}{91,000 \times 1} = 1.21 \times 10^{-6} M = \underline{1.21 \mu M}$$

2. Convert μM to mg/mL:

$$1.21 \mu M \times 0.119 \text{ mg/mL} = 0.144 \text{ mg/mL}$$

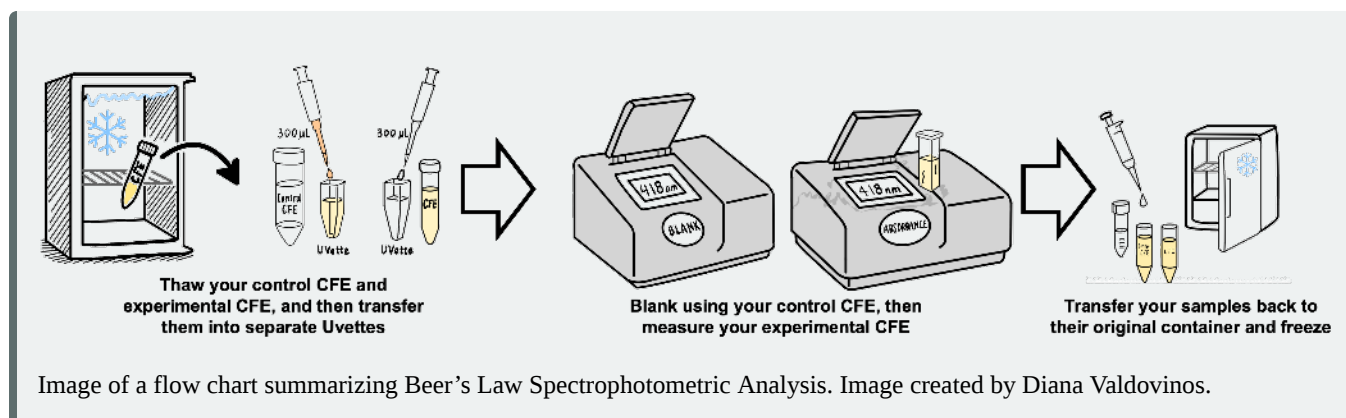
- Molecular weight of P450_{BM3} $\approx$ **119,000 g/mol**.
- $1 \mu M = 1 \times 10^{-6} \text{ mol/L} \times 119,000 \text{ g/mol} = 0.119 \text{ g/L}$
- same as 0.119 mg/mL

Therefore: **$1.21 \mu M \times 0.119 \text{ mg/mL} = 0.144 \text{ mg/mL}$**

Calculate Percent of P450 in CFE:

If the Bradford assay (From previous experiments) gave you **1.45 mg/mL total protein** in your CFE, then: % P450 = $0.144/1.45 \times 100\% = \underline{9.9\%}$

% of Total Protein	Rating	Interpretation
>20%	★ Excellent	You are getting extra credit for this!
10–20%	✓ Good	This is exactly what we need to study protein function.
5–10%	⚠ Moderate	May be usable. Maintain on ice
<5%	✗ Poor	Failed extraction.



Post-Lab Questions:

1. Absorbance Patterns:

Examine the absorbance data you collected.

- Do both CFE-A and CFE-B show a peak at **418 nm**?
- Which extract likely contains the **P450-expressing cells**?

2. Calculating Protein Concentration (Beer's Law):

Using Beer's Law ($A = \epsilon cL$), calculate the concentration of the P450 enzyme:

- Use an extinction coefficient (ϵ) of **91,000 M⁻¹cm⁻¹** for heme at 418 nm.
- Assume the **path length (L)** of the UVette is **1 cm**.
- Rearrange the formula to solve for **c (concentration)**:
 $c = A / (\epsilon L)$
- Plug in your absorbance values to determine the **molar concentration** of P450 in your sample.

Learning Objectives

After completing this lab and lecture, you should be able to:

- Apply Beer's Law to calculate protein concentration from spectrophotometric data.
- Interpret the UV-Vis absorbance spectrum of cytochrome P450 enzymes.
- Determine which samples express active P450 enzyme based on heme absorbance.
- Reflect on how buffer composition, enzyme folding, and spectrophotometric setup affect the accuracy of biochemical measurements.

3: Activity 1-3 - Beer's Law Spectrophotometric Analysis is shared under a [not declared](#) license and was authored, remixed, and/or curated by LibreTexts.

4: Activity 1-4 - Analysis of Enzymatic Activities of Cell-Free Extracts

Learning Objectives

- Understand the role of cytochrome P450 enzymes in biological systems.
- Explain the importance of NADPH in P450-catalyzed hydroxylation reactions.
- Define and apply Beer's Law in the context of spectrophotometric analysis.
- Describe how absorbance changes at 340 nm are used to track enzyme activity.
- Prepare a reaction mixture and outline the rationale behind each component.

Definition: Term

- **Cytochrome P450:** A family of heme-containing enzymes involved in oxidation reactions, often used to metabolize drugs and fatty acids.
- **P450_{BM3}:** A bacterial version of P450 that is soluble and highly active; often used in research for enzymatic activity studies.
- **NADPH (Nicotinamide Adenine Dinucleotide Phosphate, Reduced):** A cofactor that donates electrons in biosynthetic reactions; oxidized to NADP⁺ in P450 activity.
- **Beer's Law:** A relationship that connects absorbance (A) to the concentration (c) of a substance in solution: $A = \epsilon cl$.
- **Extinction Coefficient (ϵ):** A constant that indicates how strongly a substance absorbs light at a particular wavelength. For NADPH at 340 nm, $\epsilon = 6,220 \text{ M}^{-1}\text{cm}^{-1}$.
- **Specific Activity:** A measure of enzyme activity normalized to protein concentration: μmol of substrate converted per second per mg protein.
- **Spectrophotometer:** An instrument that measures how much light a sample absorbs at a particular wavelength.
- **Absorbance at 340 nm:** A measure used to monitor NADPH oxidation in real time. As NADPH is used up, absorbance decreases.

Beer's Law Spectrophotometric Analysis of P450 Enzyme Expression

In this activity, we will analyze the **enzymatic activity** of cell-free extracts that contain the cytochrome P450_{BM3} enzyme. Cytochrome P450 enzymes are a large family of heme-containing monooxygenases that play critical roles in the metabolism of a wide range of endogenous and exogenous compounds. One of the most common ways to measure the activity of these enzymes is by monitoring the **oxidation of NADPH** (nicotinamide adenine dinucleotide phosphate, reduced form), which acts as an essential cofactor in the P450 catalytic cycle.

The enzymatic activity can be assessed by measuring the change in absorbance at **340 nm** using a spectrophotometer. This wavelength corresponds to the absorbance peak of NADPH, which decreases as it is oxidized to NADP⁺ during the enzymatic reaction. Specifically, for each hydroxylation reaction catalyzed by cytochrome P450_{BM3}, one molecule of NADPH is consumed. Therefore, the rate of NADPH oxidation provides a **direct** and quantitative measure of the enzyme's catalytic activity. The molar extinction coefficient of NADPH at 340 nm is **6,220 M⁻¹cm⁻¹**, which is used to convert absorbance changes into actual concentrations and reaction rates. This spectrophotometric method provides a reliable, real-time approach to evaluate the activity of cytochrome P450 enzymes in biological samples.

Materials:

1. **Cell-free extracts (CFE)**
2. **1.5 mL centrifuge tubes**
3. **Phosphate buffer (PO₄ buffer):** 100 mM potassium phosphate, pH 7.4 (Pre-incubated at 37°C), OR 0.25X PBS
 - Made from 1L of a 1M stock solution with the following:
 - 95g of monobasic potassium phosphate (Carolina, #884250)
 - 52.5g of dibasic potassium phosphate (Carolina, #884300)
4. **Distilled water**
5. **10 mM lauric acid in 50 mM potassium carbonate** (stored at -20°C)

- Lauric acid (Carolina, #871840)
 - Potassium carbonate (Carolina, #882836)
 - Note: Make sure to **filter** the Lauric acid!
6. **10 mM NADPH in water** (stored at -20°C)
 - **Always make this fresh. This can only last up to 3 months.**
 7. **UV/VIS Spectrophotometer** (Thermo Scientific, #840-300000)
 8. **UVette cuvettes** (Eppendorf, #952010051)
 9. **Pipettes and tips**
 10. **Parafilm, Sharpie, acetone, cotton, scissors**
-

Procedure:

1. Pre-warm the following tubes at 37°C.
 - Cell-Free Extract (CFE)
 - Only take an aliquot of CFE, do not warm all of them as they will degrade over time if not frozen!
 - Phosphate buffer
 - Lauric Acid
2. Obtain a cuvette and mix the following components (refer to the table below). Do **not** add NADPH yet.
 - 50 μ L of 10 mM Lauric acid
 - 390 μ L of Phosphate buffer
 - 50 μ L of CFE
3. Blank the spectrophotometer using this mixture (From step 2) at **340 nm**.
4. Add 10 μ L of NADPH to the reaction mixture and seal it with parafilm.
5. Quickly mix the cuvette and immediately measure the initial absorbance at 0 seconds and the final absorbance at **2 minutes**.
6. **You may need to repeat this several times:** Optimize the volume of Lauric acid, CFE, and/or NADPH to achieve a decrease in absorbance between 0.2 to 0.4 after 2 minutes.
 - **Increase lauric acid concentration** (affects k_m): Providing more substrate can enhance enzyme turnover, leading to a faster NADPH oxidation and greater absorbance drop.
 - **Increase enzyme (CFE) volume** (affects V_{max}): More enzyme increases catalytic capacity, accelerating the reaction rate.

Measuring P₄₅₀-Driven NADPH Oxidation:

1. Record the Absorbance at two time points
 - For example
 - 0 second = 0.4192
 - 120 second = 0.2451
2. Calculate the differences in the two absorbance values
 - For example: $0.4192 - 0.2451 = 0.1741$
3. Convert Absorbance Change to NADPH Concentration (extinction coefficient is $6220 \text{ M}^{-1}\text{cm}^{-1}$).
 - $c = A/EL$
 - $\Delta \text{NADPH concentration} =$
4. Calculate the initial rate
 - $\text{rate} = \Delta \text{concentration} / \Delta \text{time}$
 - $28 \mu\text{M} / 120 \text{ s} = 0.233 \mu\text{M/s}$
5. Determine Protein Amount in the Reaction
 - Use Bradford assay to get protein concentration (e.g., **0.119 mg/mL**)
 - Multiply by volume added (e.g., 0.1 mL)
 - $\text{Protein amount} = 0.119 \text{ mg/mL} \times 0.1 \text{ mL} = 0.0119 \text{ mg}$
6. **Normalize** Rate to Protein
 - $\text{Specific activity} = \text{rate} / \text{protein}$

▪ $(0.233 \mu\text{M/s}) / (0.0119\text{mg}) = \underline{19.6\mu\text{M/s/mg}}$

Specific Activity ($\mu\text{M/s/mg}$)	Rating	Interpretation
>15	★ Excellent	High activity; Deserves Extra credit!
5–15	✓ Good	Useful enzyme.
1–5	⚠ Moderate	Low quality.
<0.1	✗ Very Poor	Dead protein.

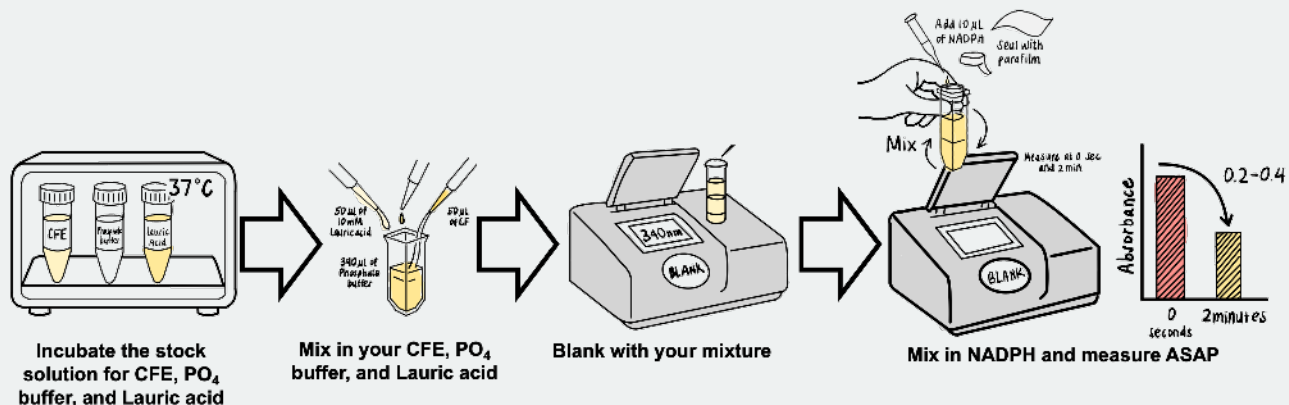


Image of a flow chart summarizing the Analysis of Enzymatic Activities of Cell-Free Extracts. Image created by Diana Valdovinos.

Post-Experiment Questions:

1. Graph the Absorbance vs. Time:

- Create a graph of absorbance over time.
- Add a trendline to the data.
- The slope of the trendline will give the **rate of the reaction** in terms of optical density (O.D.) per second.

2. Calculate the Rate of NADPH Oxidation:

- Use the following equation, derived from Beer's law: $\text{O.D.} = E \times c \times l$. $\text{O.D.} = E \times c \times l$
- Where:
 - E is the molar extinction coefficient of NADPH at 340 nm ($6220 \text{ M}^{-1}\text{cm}^{-1}$).
 - c is the concentration of NADPH.
 - l is the path length (in cm), typically 1 cm for cuvettes.
- Calculate the **rate of NADPH oxidation** in terms of **micromoles of NADPH oxidized per second**.

3. Determine the Specific Activity:

- To calculate the **specific activity** of your enzyme, express the rate of NADPH oxidation in **micromoles of NADPH oxidized per second per milligram of protein**.
- First, determine the **amount of protein** in your samples.
- Then, calculate how much **1 mg of protein** contributes to the NADPH oxidation rate.

Learning Objectives

By the end of this lab and lecture discussion, students should be able to:

- Graph and interpret real-time enzymatic data.
- Use absorbance values to calculate NADPH oxidation rates.
- Apply Beer's Law to convert absorbance into concentration units.
- Determine the specific enzymatic activity of cell-free extracts.
- Troubleshoot potential problems in enzyme assays.

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5: Activity 2-1 - Purification via Ammonium Sulfate Saturation of Cell-Free Extract

Learning Objectives

- Explain the principles of protein solubility and how it changes with salt concentration.
- Define and apply the concept of "salting out" using ammonium sulfate.
- Describe the steps involved in partial protein purification from a cell-free extract.
- Calculate the correct amount of ammonium sulfate needed for different saturation levels.
- Understand the importance of temperature control and sample labeling during enzyme purification.

Definition: Term

- **Cell-Free Extract (CFE):** A solution containing cellular components (including proteins) obtained after lysing cells and removing debris.
- **Salting Out:** A technique to precipitate proteins from a solution by increasing salt concentration, reducing protein solubility.
- **Ammonium Sulfate Precipitation:** A common method to fractionate proteins based on solubility by gradually increasing salt levels.
- **Protein Solubility:** The ability of a protein to stay dissolved in solution, influenced by factors like salt, pH, and temperature.
- **Precipitate:** The solid that forms when proteins aggregate and fall out of solution during salting out.
- **Supernatant:** The clear liquid remaining above the precipitate after centrifugation.
- **Dialysis:** A method used to remove small molecules (e.g., salts) from a protein solution using a semi-permeable membrane.
- **Centrifugation:** A technique that separates components of a mixture based on density by spinning at high speeds.

Note

- Always label your tubes with enzyme fraction, initials, and date.
- Keep samples cold at all times to prevent denaturation.
- Add ammonium sulfate **slowly and gently**—foaming = protein damage!
- Record your weights and volumes carefully—this ensures accurate calculations.
- Keep track of which fraction (0–30%, 30–60%, 60–100%) you're working with.

Activity 2-1: Ammonium Sulfate Saturation of Cell-Free Extract Containing Cytochrome P450_{BM3}

In this activity, you will begin the process of purifying the enzyme **Cytochrome P450_{BM3}** from a crude cell-free extract using a method called **ammonium sulfate precipitation**, or more commonly, "**salting out**." This is a foundational protein purification technique that takes advantage of how protein solubility changes in the presence of salt. Think of proteins in a solution like people at a party: when the environment is balanced, everyone moves freely and comfortably (soluble). But if the space becomes too crowded or uncomfortable—say, by adding too much salt—it disrupts the atmosphere, and some people (proteins) will leave the party (precipitate out). In technical terms, proteins are more soluble at low or moderate salt concentrations due to the stabilization of charges on their surfaces (this is called "**salting in**"). However, as salt concentration increases further, the salt starts to "hog" the water molecules, leaving fewer available to keep proteins dissolved. This causes proteins to aggregate and fall out of solution ("**salting out**").

Each protein has its own unique solubility profile depending on the type and amount of salt, pH, and temperature. By slowly increasing the concentration of **ammonium sulfate**, a highly soluble salt, you can selectively precipitate different proteins at different stages. This helps separate the target enzyme, Cytochrome P450_{BM3}, from other proteins in the mixture. Ammonium sulfate is especially useful for this kind of purification because:

- It is very soluble in water (up to 4.1 M at 20°C).
- It creates a high ionic strength environment needed for salting out.
- It often stabilizes proteins, which is why purified proteins are commonly stored in ammonium sulfate solutions.

However, a key drawback is that **the presence of salt can interfere with enzyme activity assays**. High salt concentrations can distort assay results or reduce the activity of some enzymes. Therefore, after salting out, we must **remove the excess salt** from the protein fraction before any further analysis. To do this, we use a technique called **dialysis**. Dialysis works similarly to a tea bag in water: small molecules (like salt) can pass through the bag, but larger molecules (like proteins) cannot. We seal the protein solution inside a **semi-permeable membrane (dialysis tubing)** and submerge it in a salt-free buffer. Over time, the salt diffuses out of the tubing into the surrounding buffer, leaving the proteins behind in a salt-free solution.

⚠ Important Note: Throughout the purification process, especially when collecting enzyme fractions, always keep your samples **on ice**. Many enzymes, including P450_{BM3}, are sensitive to temperature and can lose activity if left at room temperature for too long.

Objective: To partially purify the enzyme **Cytochrome P450_{BM3}** from a **cell-free extract (CFE)** by using **ammonium sulfate precipitation**. This process helps separate proteins based on their solubility.

Materials

- Cell-free extract (CFE) containing cytochrome P450_{BM3} (from previous lab)
- 1.5 mL centrifuge tubes
- 50 mL conical centrifuge tubes (Corning™ 352070)
- Pipettes and tips
- Extraction buffer: 50 mM potassium phosphate, pH 7.4 OR 0.25X PBS
 - (Prepared from 1M stock: 95 g monobasic KH₂PO₄ + 52.5 g dibasic K₂HPO₄ in 1L water)
- Dry ammonium sulfate powder (Flinn Scientific #A0062)
- 50 mL and 1L glass beakers
- Stir bar and magnetic stirrer
- Ice and ice bucket
- Weighing boats and electronic balance
- Centrifuge capable of 5000 rpm

📘 IMPORTANT! Keep all enzyme samples **on ice** throughout the procedure to prevent enzyme degradation. Loss of activity can occur if enzymes warm up.

INITIAL CONCENTRATION OF AMMONIUM SULFATE, % SATURATION	FINAL CONCENTRATION OF AMMONIUM SULFATE, % SATURATION																
	10	20	25	30	33	35	40	45	50	55	60	65	70	75	80	90	100
GRAMS SOLID AMMONIUM SULFATE TO BE ADDED TO 1 LITER OF SOLUTION																	
0	56	114	144	176	196	209	243	277	313	351	390	430	472	516	561	662	767
10		57	86	118	137	150	183	216	251	288	326	365	406	449	494	592	694
20			29	59	78	91	123	155	189	225	262	300	340	382	424	520	619
25				30	49	61	93	125	158	193	230	267	307	348	390	485	583
30					19	30	62	94	127	162	198	235	273	314	356	449	546
33						12	43	74	107	142	177	214	252	292	333	426	522
35							31	63	94	129	164	200	238	278	319	411	506
40								31	63	97	132	168	205	245	285	375	469
45									32	65	99	134	171	210	250	339	431
50										33	66	101	137	176	214	302	392
55											33	67	103	141	179	264	353
60												34	69	105	143	227	314
65													34	70	107	190	275
70														35	72	153	237
75															36	115	198
80																77	157

Image of Ammonium Sulfate table. Table created by Diana Valdovinos.

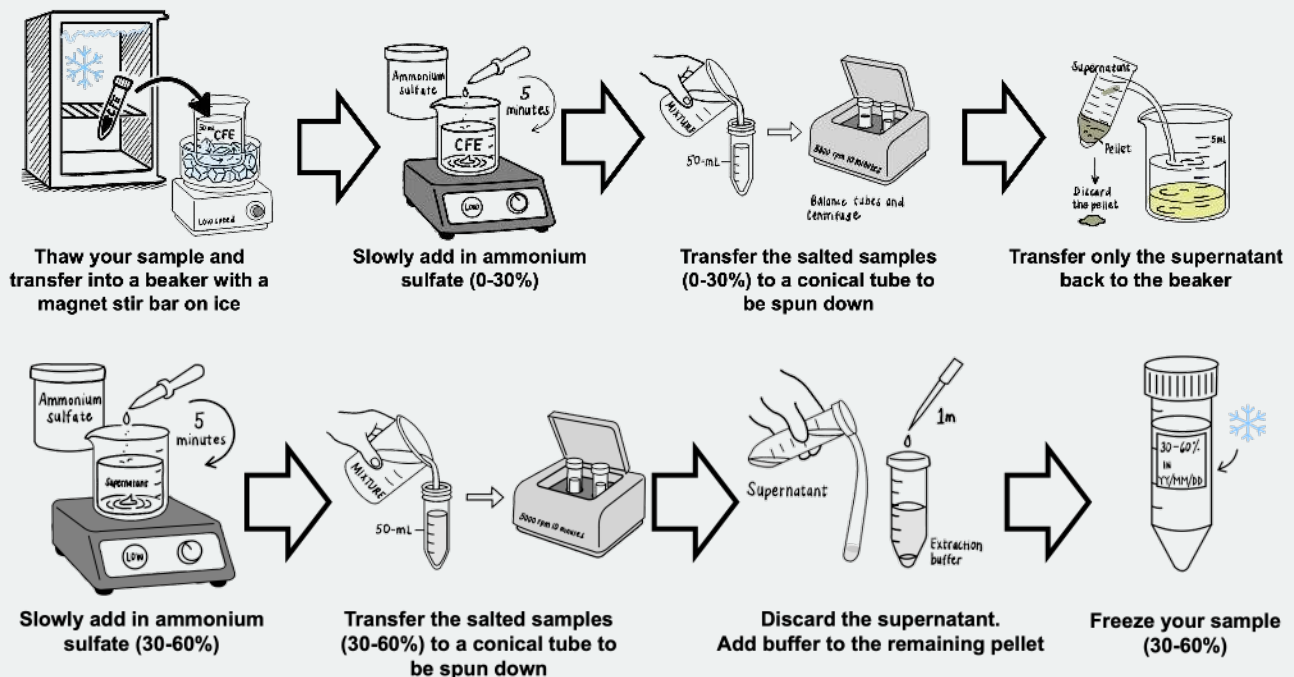


Image of a flow chart summarizing the Purification via Ammonium Sulfate Saturation of Cell-Free Extract. Image created by Diana Valdovinos.

Protocol

Initial Set Up

1. **Thaw** your CFE, and estimate the volume of your sample by eyes.
 1. *Thaw on ice is preferred, but if you are in a rush, you can thaw by hand. However, store on ice afterwards.*
2. Using a pipette, transfer **50 µL** of CFE into a **1.5 mL centrifuge tube**.
 - Label as: "**CFE-SDS**" and your **initial and date**
 - Freeze this aliquot.
3. Place a **clean stir bar** in a 50-mL beaker, and then pour in your CFE.
4. Place this 50-mL beaker inside a **1L beaker filled with ice** to keep it cold.
5. Start the magnetic stirrer on **low speed**.
 - *Avoid any bubbles as it indicates protein denaturation*

You'll add ammonium sulfate in **three steps**:

- 0–30% (Mostly small, highly soluble proteins)
- 30–60% (P450 BM3 is commonly enriched here)
- 60–100% (Proteins with very high solubility precipitate)

Each step isolates different protein fractions.

Step 1: 0–30% Saturation

1. **Calculate** the amount of ammonium sulfate needed.
 - Use a reference table or formula. Example:
 - For 10 mL of CFE, the table says 176 g/L is needed. → **Add 1.76 g** ammonium sulfate.
2. Slowly **add ammonium sulfate** in small portions while stirring gently. As salt is added, the solution will turn **cloudy**—this means proteins are precipitating.
3. After the final addition, stir for **5 more minutes**. Avoid foam (indicates denaturation).

4. Transfer the mixture to a **50 mL conical tube**.
 5. **Balance** tubes and **centrifuge** your mixture at 5000 rpm for 10 minutes.
 6. Clean out your 50-mL beaker.
 7. Slowly transfer only the supernatant back into the clean 50-mL beaker.
 8. **Discard the pellet.**
-

Step 2: 30–60% Saturation and 60–100% Saturation

1. Repeat the process and calculate the amount of ammonium sulfate needed to increase from 30% to 60%.
 - Example: If the table says 198 g/L → For 10 mL, then add **1.98 g**.
 2. **Slowly** add ammonium sulfate to your supernatant with gentle stirring.
 3. Stir for 5 more minutes after the final addition.
 4. Transfer the mixture to a clean 50-mL conical tube and centrifuge at 5000 rpm for 10 min.
 5. **Discard the remaining supernatant.**
 6. Resuspend the pellet in **1 mL of extraction buffer**.
 7. Label and freeze as: "**30–60% - initial, [date]**"
-

Learning Objectives

After completing this activity, you should be able to:

- Justify the use of ammonium sulfate for partial protein purification.
- Interpret results from each precipitation step and predict where your target protein might be found.
- Troubleshoot common errors in salting out and propose corrective steps.
- Explain how dialysis is used to prepare samples for further enzymatic assays.

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6: Activity 2-2 - Dialysis for Enzyme Purification

Learning Objectives

- Explain the purpose of dialysis in enzyme purification.
- Describe how a semipermeable membrane works in dialysis.
- Identify what components are retained vs. removed during dialysis.
- Understand how ammonium sulfate affects protein solubility and why its removal is necessary.
- Predict how impurities like salts may interfere with downstream assays.

Definition: Term

- **Dialysis:** A method used to remove small molecules (e.g., salts) from a solution containing larger macromolecules (e.g., proteins), using a semi-permeable membrane.
- **Semi-permeable Membrane:** A material with pores small enough to allow passage of small molecules while retaining larger ones.
- **Ammonium Sulfate Saturation:** A method to precipitate proteins based on their solubility in high salt concentrations.
- **Cell-Free Extract (CFE):** A lysate from cells where all cellular structures are broken down, leaving behind soluble proteins and other molecules.
- **Extraction Buffer:** A solution that maintains protein stability during purification by controlling pH and ionic strength.
- **Bradford Assay:** A colorimetric assay used to measure total protein concentration using Coomassie Brilliant Blue dye.
- **Beer's Law ($A = \epsilon cl$):** A formula that relates absorbance to concentration, path length, and molar extinction coefficient—used to quantify specific proteins like P450.

Dialysis for Enzyme Purification

In this lab, we are continuing the purification of the P450 enzyme from a cell-free extract using a technique called **dialysis**. At this stage, the enzyme has already been partially purified using **ammonium sulfate saturation**, which helps precipitate out proteins based on their solubility. However, the remaining solution still contains excess ammonium sulfate and other small molecules that we need to remove before we can further analyze the enzyme.

Dialysis works like a molecular sieve or a "**tea bag for molecules**." Imagine placing a mixture of small and large molecules into a special bag made of a semi-permeable membrane—this membrane has tiny pores that allow small molecules like salts and buffer components to pass through while trapping larger molecules like proteins inside. When you submerge the bag in a clean buffer solution, the small molecules diffuse out, but the proteins stay in. By repeatedly changing the surrounding buffer, we gradually remove unwanted small molecules while keeping our enzyme safe and contained.

This method is essential because the presence of salts like ammonium sulfate can interfere with downstream applications such as protein concentration measurements or enzyme activity assays. After dialysis, your enzyme samples will be in a clean extraction buffer and ready for quantification and functional testing.

Materials:

- Pipettes and tips
- Frozen protein fractions (from previous lab)
- Dialysis tubing (soaked in ddH₂O in a 100 mL beaker)
- Dialysis clips (Carolina #684216 and #684239)
- 2 L beaker containing 1 L cold Extraction Buffer with a magnetic stirrer
- **COLD Extraction Buffer:** 50 mM potassium phosphate, pH 7.4 OR 0.25X PBS
 - Prepared from from 1 M stock:
 - 95 g of monobasic potassium phosphate (Carolina #884250)
 - 52.5 g of dibasic potassium phosphate (Carolina #884300)

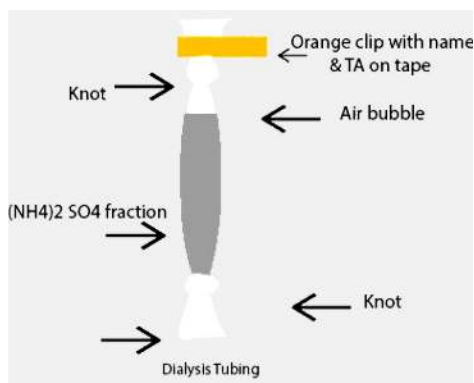


Image created by Diana Valdovinos.

Procedure:

1. **Thaw frozen 30-60% fractions** from the previous lab completely on ice.
2. Take and prepare a piece of dialysis tubing from the water-filled beaker.
 - Tie one end securely with a dialysis clip or a tight knot.
 - Carefully pipette your protein fraction into the open end of the tubing.
 - Remove any air bubbles by gently squeezing the tubing.
 - Tie off the open end and secure it with a second dialysis clip.
 - Label the clip or tubing clearly for identification.
3. Place the sealed dialysis tubing with your samples in a **2 L beaker** containing **1 L of cold Extraction Buffer** on a magnetic stirrer.
4. Let it stir gently for **1 hour**.
5. After dialysis, transfer the contents of each tubing into a clean, **labeled 1.5 mL centrifuge tube**. Label as "Dialysis, initial, date"
6. Then transfer a **50 µL aliquot** into another clean tube and label as "Dialysis-SDS, initial date"
7. Store the samples in a freezer until further use.

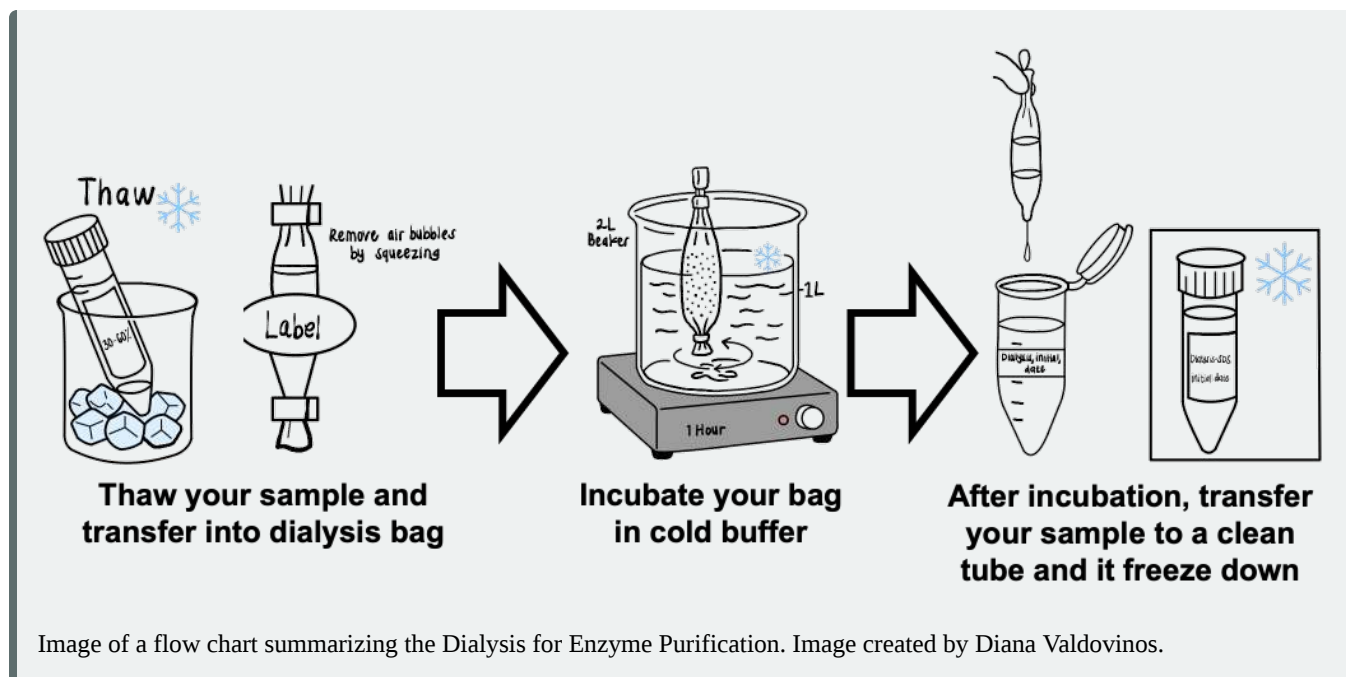


Image of a flow chart summarizing the Dialysis for Enzyme Purification. Image created by Diana Valdovinos.

Learning Objectives

After completing the lab and associated analysis, students should be able to:

- Perform dialysis to purify an enzyme sample.
- Explain the outcome of dialysis based on visual and analytical results.
- Accurately measure protein concentration using the Bradford assay.
- Use Beer's Law to determine P450 concentration from absorbance data.
- Evaluate enzyme purity and functionality through specific activity calculation.
- Interpret purification data and conclude each fraction's quality.

Key Takeaways:

- Dialysis is a simple yet powerful technique that removes contaminants without harming the protein.
- Buffer composition and temperature matter in preserving enzyme structure and activity.
- Proper labeling and sample handling are essential to track purification progress.
- Quantitative techniques like the Bradford assay and Beer's Law provide insight into the effectiveness of purification.
- Comparing fractions allows identification of the most enriched and active enzyme preparation.

Note

Next Lab Tasks: We will use the the dialyzed fractions to perform the following analyses:

- Measure total protein concentration using the Bradford Assay.
- Quantify P450 concentration using Beer's Law ($A = \epsilon cl$).
- Determine enzyme activity (Rate of reaction in $\mu\text{M}/\text{sec}/\text{mg}$ protein).

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7: Activity 2-3 - Size Exclusion Gel Filtration Purification of Enzymes

Learning Objectives

- Define gel filtration chromatography and explain how it separates molecules by size.
- Describe the physical structure and function of Sephadex G-25 beads.
- Predict elution order based on protein size in the context of an exclusion limit.
- Understand the purpose of this technique in the multi-step purification of Cytochrome P450_{BM3}.
- Identify materials and prepare the gel filtration column correctly.

Definition: Term

- **Chromatography:** A technique for separating mixtures based on physical or chemical properties like size, charge, or affinity.
- **Gel Filtration Chromatography:** A type of chromatography that separates molecules based on size using porous beads; also called size-exclusion chromatography.
- **Exclusion Limit:** The maximum molecular weight that can enter the pores of a gel bead; molecules larger than this size are excluded and elute faster.
- **Sephadex G-25:** A type of gel filtration matrix made from dextran beads with pores that exclude proteins larger than ~25 kDa.
- **Elution:** The process by which molecules exit the chromatography column and are collected in fractions.
- **Cytochrome P450_{BM3}:** A heme-containing bacterial enzyme, often red-brown in color, used here as a model for protein purification studies.

Key Concepts

- Molecular weight (kDa) and how it affects protein diffusion.
- Previous lab techniques: ammonium sulfate precipitation, dialysis, Bradford assay, and Beer's Law.
- Why it's important to purify proteins in biotechnology and research settings.

Gel Filtration Chromatography

Chromatography is a powerful technique used to separate complex mixtures of biological molecules based on their physical and chemical properties. In this activity, we will focus on **gel filtration chromatography**—a type of chromatography that separates proteins **based on their size**. This method is also called **size-exclusion chromatography**. Previously, you began purifying the enzyme **Cytochrome P450_{BM3}** from *E. coli* by using ammonium sulfate precipitation, which allowed you to separate some soluble proteins based on solubility. You then dialyzed those protein fractions to remove small molecules like salts. Now, we'll use gel filtration to take the purification process a step further.

Gel filtration uses a column packed with tiny porous beads made of a polymer like **Sephadex**. Think of these beads as **sponges with tunnels inside them**. When a mixture of proteins is poured onto the column, the proteins take one of two general routes:

- **Large proteins**, like Cytochrome P450_{BM3}, are **too big** to fit into the pores of the beads. Imagine a beach ball rolling over a sponge—it doesn't sink in, so it moves around the beads and flows through the column **faster**.
- **Small proteins**, on the other hand, **enter the pores**—like marbles sinking into the sponges—so they take a longer, winding path through the beads and come out **later**. This is why **larger molecules elute (come out) first**, and **smaller ones elute later**.

Common misconception: In gel filtration chromatography, molecules are separated by **size**, but not in the way students often assume. The **Sephadex G-25 column** used here has an **exclusion limit of 25 kDa**. Molecules **smaller than 25 kDa** can enter the pores in the gel beads, so they take a longer, winding path through the column and **elute later**. Molecules **larger than 25 kDa** are **excluded from the pores** and take a more direct route around the beads, **eluting earlier**. Now, here's where it gets interesting: while both **60 kDa** and **120 kDa** proteins are too large to enter the beads, the **120 kDa protein elutes slightly faster** than the 60 kDa one. This is because larger molecules interact **even less** with the gel and diffuse through the column more directly. So yes, **both are excluded**, but the **largest molecules still come out first**. So, in gel filtration, **the more excluded a molecule is, the faster it elutes**. The biggest ones race out first—not because they're fast, but because they're **too big to get delayed**.

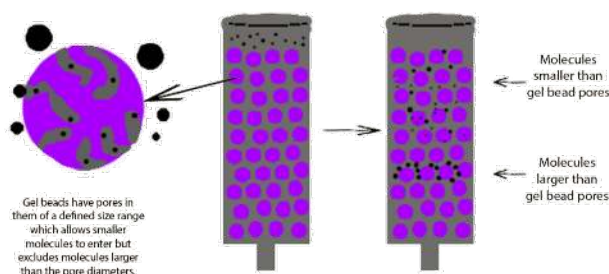


Image created by Diana Valdovinos.

Materials Needed

- Make sure you have all materials and protocol from your previous labs on:
 - **Bradford Assay**
 - **Beer's Law**
 - **Enzyme Kinetics**
- Gel filtration column with stopcock
- Sephadex-G25 beads (Sigma-Aldrich, #G25150-50G)
- **Dialyzed reddish sample** from the previous lab
- **Equilibration buffer**: 100 mM potassium phosphate, pH 7.4 OR 0.5X PBS
 - Prepared from 1 M stock solution:
 - 95 g monobasic potassium phosphate (Carolina, #884250)
 - 52.5 g dibasic potassium phosphate (Carolina, #884300)
- 1.5 mL microcentrifuge tubes
- 15 mL conical centrifuge tube
- Waste container (for collecting flow-through from the column)
- Ice bucket for keeping fractions cold

Procedure

Prepare Your Gel Filtration Column

1. Weigh out **1 g of Sephadex-G25 beads**.
2. Add the beads to **10 mL of 100 mM potassium phosphate buffer** in a 15 mL conical tube.
3. Gently **invert the tube continuously for 5 minutes** to fully hydrate and mix the beads.
4. Carefully pour the bead suspension into the **gel filtration column** until it's about **half-full (~50%)**.
5. Let the buffer drain slowly through the column, allowing the beads to **settle and pack**. Once settled, **close the stopcock**.

Run Your Sample

1. Take your **dialyzed protein sample** from the freezer and allow it to **thaw on ice**.
2. Gently pipette the entire **sample** onto the top of the column matrix.
3. Wait for the **meniscus to reach the top of the settled bead bed**, then **close the stopcock**.
4. Discard the liquid wastes.
5. Add **300 μ L of buffer at a time** to the top of the column, and collect the liquid that comes out into a **Cuvette tube**.
6. Measure and record your solution at 418nm via Spectrophotometry Beer's Law method.
 - **Cytochrome P450_{BM3}** has a natural red-brown heme color that absorbs at 418nm.
7. Repeat this step (3-5) until the absorbance via Beer's Law reading is low.
8. Combine the **entire volume** of the top 3 solutions with the highest absorbance into a single microcentrifuge tube.
9. You may freeze down your samples or continue to the next QA and QC steps (Bradford, Beer's, and Kinetics)

Congratulations! You've successfully completed gel filtration purification of Cytochrome P450_{BM3}.

1. Measure and **record the total volume** of this combined fraction.
2. Take a **50 μ L aliquot** of your purified fraction, transfer it into a new labeled microcentrifuge tube, and freeze it.

- Label the tube as: "Pure WB - initial, [Your Name], [Date]"

3. Analyze Your Purified Enzyme:

- Determine total protein concentration** using the **Bradford Assay**.
- Measure Cytochrome P450_{BM3} concentration** using **Beer's Law** (based on absorbance at the Soret peak).
- Measure the enzymatic activity** of P450_{BM3} using your kinetic assay method.

4. Create a **data table**. Record your observations, volumes, concentrations, and calculations

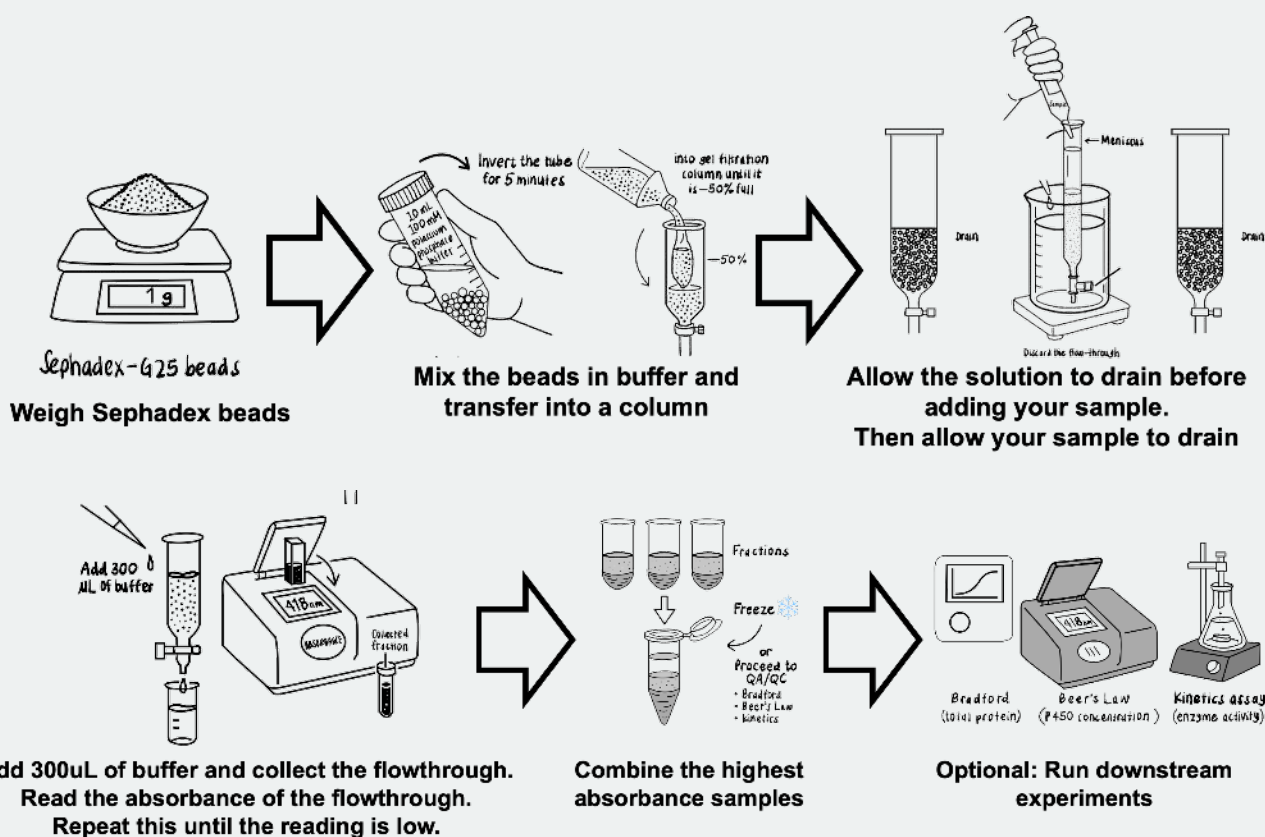


Image of a flow chart summarizing the Size Exclusion Gel Filtration Purification of Enzymes. Image created by Diana Valdovinos.

Data Table:

Fraction	Protein conc. (µg/µL)	P450 conc. (µM)	Rate (µM/s/mg)
CFE A	(From previous results)	(From previous results)	(From previous results)
CFE B	(From previous results)	(From previous results)	(From previous results)
30–60% (after dialysis and filtered)			

Learning Objectives

After completing the activity, you should now be able to:

- Describe how gel filtration separates proteins by size and interpret real elution data.
- Identify which fractions contain your protein of interest based on color and elution order.
- Explain why larger proteins elute faster in gel filtration, despite intuition suggesting smaller should move faster.

- Analyze and report protein concentration, purity, and enzymatic activity after purification.
- Integrate gel filtration as one step in a broader multi-technique protein purification strategy.

7: Activity 2-3 - Size Exclusion Gel Filtration Purification of Enzymes is shared under a [not declared](#) license and was authored, remixed, and/or curated by LibreTexts.

8: Activity 2-4 - Verification of Protein Purification via Polyacrylamide Gel Electrophoresis

Learning Objectives

- Define the purpose of PAGE and explain its role in protein analysis.
- Describe how SDS-PAGE separates proteins by size.
- Explain the function of key reagents such as SDS, β -mercaptoethanol (BME), and polyacrylamide.
- Identify the components and layers of a polyacrylamide gel.
- Predict how protein size influences migration in PAGE.
- Understand how to prepare, load, and run protein samples in a vertical gel system.

Definition: Term

- **Polyacrylamide Gel:** A synthetic gel matrix used in electrophoresis to separate proteins based on size and charge.
- **SDS (Sodium Dodecyl Sulfate):** An anionic detergent that denatures proteins and gives them a uniform negative charge.
- **β -Mercaptoethanol (BME):** A reducing agent that breaks disulfide bonds, helping proteins unfold.
- **Denaturation:** The process of unfolding proteins by disrupting their secondary and tertiary structures.
- **Stacking Gel (4%):** The top layer of the gel with large pores, which compresses proteins into a tight band.
- **Resolving Gel (10%):** The bottom layer with smaller pores that separates proteins by size.
- **Electrophoresis:** A technique that uses an electric field to move molecules through a gel.
- **Protein Ladder (Standard):** A mixture of proteins with known molecular weights used as a size reference.
- **Bromophenol Blue:** A small, visible dye that tracks the progress of electrophoresis without indicating protein size.
- **ChemiDoc Imager:** A digital imaging system that visualizes protein bands on stain-free gels.

Polyacrylamide Gel Electrophoresis (PAGE)

Background:

Polyacrylamide Gel Electrophoresis (PAGE) is a widely used technique in molecular biology and biochemistry to separate proteins based on their size and charge. In this activity, you will apply PAGE to evaluate the purity of your protein samples—specifically the P450_{BM3} enzyme—by observing their migration through a polyacrylamide gel matrix. Each student group will run their own gel and visualize results using a **ChemiDoc Imaging System**, which enables quick, **stain-free** detection of protein bands, eliminating the need for traditional staining and de-staining steps using Coomassie blue.

PAGE separates proteins by exploiting their differential movement through a gel when subjected to an electric field. Proteins migrate at different rates depending on several factors: the gel's **porosity**, the pH and ionic strength of the buffer, temperature, and the protein's **size and charge**. In standard SDS-PAGE (Sodium Dodecyl Sulfate-PAGE), proteins are first denatured and coated with SDS, an anionic detergent. SDS disrupts secondary and tertiary structures, causing the proteins to unfold into linear shapes. It also imparts a uniform negative charge across the length of each polypeptide, ensuring that protein migration is primarily influenced by size—not by their intrinsic charge or shape. To further disrupt protein folding and break disulfide bonds, the sample buffer includes β -mercaptoethanol (BME) and the samples are boiled before loading onto the gel.

The gel itself consists of polyacrylamide, a synthetic polymer formed by cross-linking acrylamide and bisacrylamide. Polymerization is initiated by ammonium persulfate (APS), an oxidizing agent, and catalyzed by TEMED, which accelerates APS decomposition. The gel is cast in two layers:

1. **The stacking gel** (4% acrylamide, upper layer) has large pores and functions to concentrate all proteins into a narrow band, improving resolution.
2. **The resolving (or separating) gel** (10% acrylamide, lower layer) has smaller pores and separates proteins based on size during electrophoresis.

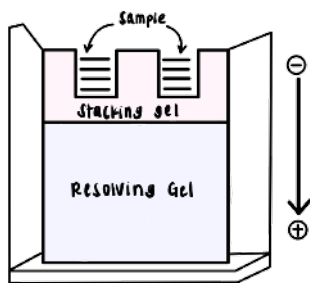


Image created by Diana Valdovinos.

Your gel will be run in a vertical electrophoresis system. This apparatus has two buffer chambers separated by the gel, with electrodes connected to a power supply (black = cathode, red = anode). When an electric current is applied, the negatively charged SDS-protein complexes migrate through the gel towards the positive anode. Smaller proteins move more quickly through the matrix, while larger proteins experience more resistance and migrate more slowly. For reference, Cytochrome P450_{BM3} has a molecular weight of approximately 119 kDa.

To monitor the progress of electrophoresis in real time, the loading buffer contains a tracking dye—**bromophenol blue**. This small dye molecule runs ahead of the proteins and is not used to determine size but gives a visual cue for when to stop the gel run. Once the dye front approaches the bottom of the gel, the electrophoresis can be halted.

After the run, you will use a **ChemiDoc Imager**, a high-resolution imaging system that allows you to detect and document your protein bands directly. By comparing the migration distance of your bands to a known **Protein Standard ladder** (which contains proteins of known molecular weights), you can estimate the molecular weights of your sample proteins and assess their purity. A single, distinct band at the expected molecular weight indicates a successful purification, while multiple bands suggest the presence of contaminants or incomplete purification.

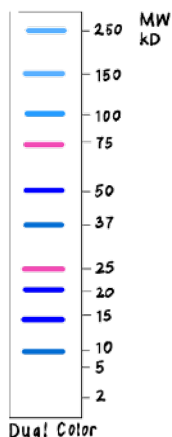


Image created by Diana Valdovinos.

This activity will help you validate whether your target protein was successfully purified and assess the integrity of the enzyme P450_{BM3} as part of your larger experimental workflow.

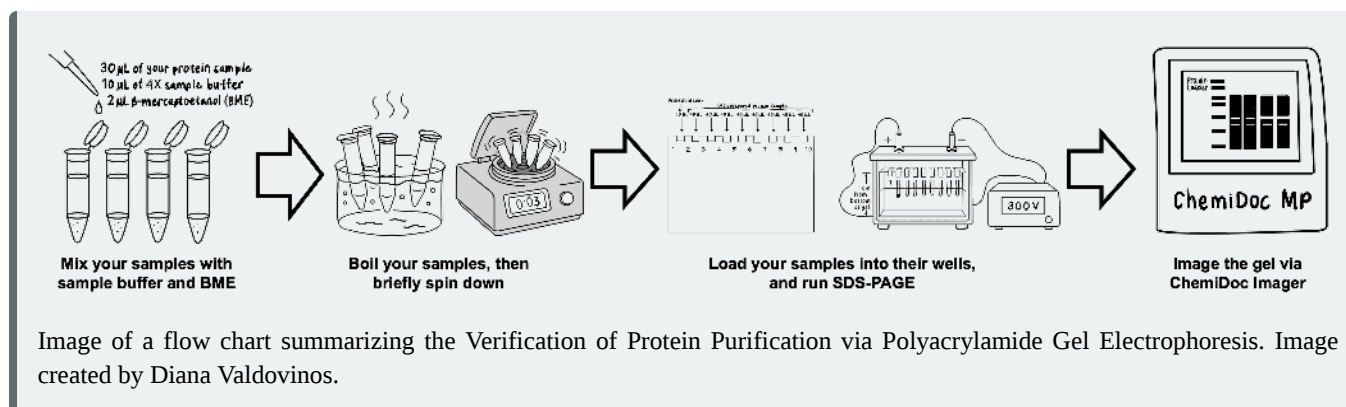
Materials Needed

- **Electrophoresis system with power supply** (Bio-Rad #1658005 and #1645050)
- **10% TGX stain-free polyacrylamide gels** (Bio-Rad #4568034)
- **10X Running buffer** (Bio-Rad #1610772)
- **4X Sample buffer** (Bio-Rad #1610747)
- **Protein samples** (from previous labs):
 - CFE A, CFE B, 30–60% (Dialyzed), and 30–60% (PURE)
- **Protein standard (ladder)** (Bio-Rad #1610363)
- **Heat source** (Boiling water or 100°C heating block – Carolina #217200 or #217205)

- Micropipettes and pipette tips
- 1.5 mL microcentrifuge tubes
- Gel opening lever (Bio-Rad #4560000)
- ChemiDoc MP Imaging System (Bio-Rad #12003154)
- β -mercaptoethanol (BME)

Procedure: Preparing and Running Your Gel

1. Take new 1.5 mL microcentrifuge tubes and label one for each protein sample. Include your group name on the labels.
2. In each labeled tube, combine:
 - 30 μ L of your protein sample
 - 10 μ L of 4X sample buffer (final concentration will be 1X)
 - 2 μ L of β -mercaptoethanol (BME)
3. Cap the tubes tightly and place the tubes in **boiling water** or a **95°C heating block** for **4 minutes** to denature the proteins and reduce disulfide bonds.
4. Let the tubes cool at **room temperature for 1 minute**.
5. Briefly **spin down** the tubes (3 seconds) in a microcentrifuge to collect any condensation from the lid.
6. Refer to the sample loading diagram provided below.
7. **Your TA or instructor will load 10 μ L of the protein standard into well 1.**
8. Carefully and slowly load **40 μ L** of each prepared sample into separate wells on the gel.
9. Place the lid securely on the electrophoresis tank.
10. Set the power supply to **300 volts**.
11. Run the gel until the **bromophenol blue tracking dye** is approximately **1 cm from the bottom** of the gel.
12. Use the **gel-opening lever** to open the gel cassette.
13. Place the gel in the **ChemiDoc MP Imaging System** to view and document your protein bands.



Learning Objectives

By the end of this activity, you should be able to:

- Interpret protein band patterns to assess purity and estimate molecular weight.
- Evaluate your experimental technique and identify potential loading or handling errors.
- Compare experimental bands to the protein ladder to determine approximate sizes.
- Troubleshoot common issues such as smeared bands or incomplete denaturation.
- Reflect on how PAGE supports broader experimental workflows in protein purification and analysis.

Self-Evaluation

Rate your confidence on the following skills from 1 (Needs Work) to 5 (Confident):

Skill	Rating (1–5)
Preparing protein samples with BME and sample buffer	☐
Correctly loading the gel without introducing bubbles	☐
Running the electrophoresis at appropriate voltage	☐
Identifying molecular weights using a protein ladder	☐
Operating the ChemiDoc Imager and capturing gel images	☐

8: Activity 2-4 - Verification of Protein Purification via Polyacrylamide Gel Electrophoresis is shared under a [not declared](#) license and was authored, remixed, and/or curated by LibreTexts.

9: Activity 3-0 - Project Instructions - Investigating Enzyme Kinetics

Background: Molecular Analysis of P450 Enzyme Reaction Rates

Enzymes are biological catalysts that accelerate chemical reactions. One of the most important ways to understand how enzymes work is by studying their **kinetics**—how fast they convert substrates into products under different conditions. A few key terms will guide your understanding of enzyme kinetics:

- **V_{max}** is the maximum rate of a reaction when the enzyme is saturated with substrate.
- **K_m** (Michaelis-Menten constant) is the substrate concentration at which the reaction rate is half of V_{max}. It provides insight into the enzyme's affinity for the substrate—a lower K_m means higher affinity.
- **K_{cat}** (turnover number) is the number of substrate molecules an enzyme can convert into product per unit time when fully saturated.

In this project, you will explore the enzyme **Cytochrome P450_{BM3}**, which plays an important role in fatty acid metabolism. In the presence of **NADPH** and **oxygen**, P450_{BM3} hydroxylates long-chain fatty acids like **lauric acid**. It uses several cofactors—**FAD**, **FMN**, and **heme**—to transfer electrons from NADPH to the enzyme's active site. The enzyme follows a classic reaction mechanism: $E + S \rightleftharpoons ES \rightarrow E + P$, where E is the **enzyme**, S is the **substrate**, ES is the **enzyme-substrate complex**, and P is the **product**. The **rate of the reaction** depends on how much enzyme and substrate are available. Early in the reaction, when substrate is abundant and product levels are low, we can focus on the **initial reaction rate**—the steepest, linear portion of the product vs. time graph.

As the substrate concentration increases, so does the rate—until the enzyme becomes saturated. At this point, further increases in substrate no longer affect the rate. This behavior is described by the **Michaelis-Menten equation**: $v = V_{\max}[S] / K_m + [S]$

This equation helps us estimate kinetic parameters from experiments. However, determining **V_{max}** from a graph is tricky since it requires an impractically high amount of substrate. A more convenient way to estimate kinetic values is to use the **Lineweaver-Burk plot**, a double reciprocal plot that provides a straight-line graph where V_{max} and K_m can be directly determined from the y- and x-intercepts.

In this project, you will analyze the activity of P450_{BM3} by exploring how substrate concentration, enzyme concentration, and inhibitors affect the enzyme's rate of reaction. You will also examine whether certain compounds act as **competitive** or **non-competitive inhibitors**.

Project Title: Molecular Analysis of P450 Enzyme Kinetics

Objective:

You will work in a group of 3–4 students to investigate the enzyme activity of **Cytochrome P450_{BM3}**, focusing on **reaction kinetics** and the effect of inhibitors (Imidazole and 4-Phenylimidazole) using **lauric acid** as the substrate. Your goal is to analyze whether these compounds act as **competitive or non-competitive inhibitors** and present your findings in the form of a **scientific research poster**.

Timeline:

- This is a four-week project. You will plan, perform, and analyze your experiments, culminating in an **oral poster presentation** of your findings.

Overview:

You will apply multiple laboratory techniques discussed in class, including:

- Protein quantification using the **Bradford assay**
- Absorbance measurements using **Beer's Law**
- Enzyme **kinetics analysis**
- **Data analysis**

Project Flow:

1. **Form a Group** of 3–4 students.
2. **Select a Research Question:**

- **Option 1:** Does **Imidazole** act as a competitive or non-competitive inhibitor of P450_{BM3}?
 - **Option 2:** Does **4-Phenylimidazole** act as a competitive or non-competitive inhibitor of P450_{BM3}?
3. Create a **flowchart** of your experiment.
- Define your **experimental variables** and **controls**.
 - Decide how you will **measure the initial rate of reaction**.
4. **Perform the Experiment:**
- Collect data using proper lab techniques.
 - Measure product formation over time.
 - Use graphs to determine **initial velocities**, **V_{max}**, and **K_m**.
5. **Data Analysis:**
- Use both **Michaelis-Menten** and **Lineweaver-Burk** plots.
 - Compare rates with and without inhibitors to determine the **type of inhibition**.
 - Interpret your data in the context of enzyme-substrate interaction.
6. **Poster Presentation:**
- Prepare a clear and engaging scientific poster.
 - Include sections on background, methods, results (graphs/tables), and conclusions.
 - Present your poster to the class and explain how your data supports your research question.

What to Include in Your Poster

Your poster must contain the following sections with **organized, visually clear, and well-labeled figures and tables**:

Title and Group Info (8 points)

- A clear, specific project title
- Names of group members
- School name and sponsorships
- Include any scientific papers, lab manuals, or sources you used (at least 2–3 references, APA or MLA format).
- Overall looks and presented in a professional manner

Background/Objective (5 points)

- (1pts) What were your research question(s) and the goal of your project?
- (1pts) Why is **Cytochrome P450_{BM3}** important in research, clinics, and applications?
- (1pts) What were the **K_m** and **V_{max}** for your substrate and inhibitor?
- (1pts) What type of inhibitor was your drug?
- (1pts) How would the type of inhibition be relevant or significant in clinical and real-world applications?

Methods (5 points)

- (2pts) Explain the list of experiments you did via the use of a **flowchart** to summarize each test (I love cartoon images):
 - (3pts) Choose three key experiments, and explain what were your control groups and why,

Results (39 points)

- **Figure 1. Protein Expression (From Project 3)**
 - **(2pts) Figure 1a - Scatter plot from Bradford standard curve** (Showing R² and equation)
 - (1pts) Explain what R² represents
 - (1pts) What was the concentration of your CFE (mg/mL)?
 - **(2pts) Figure 1b - Pie Chart from Beer's Law** (Percent of your P450_{BM3} is in CFE)
 - (1pts) What was the concentration of your p450_{bm3}?
- **Figure 2. Determining the K_m for enzyme and substrate**
 - **(2pts) Figure 2a - Line graph from kinetics of different Enzymatic concentrations** (Enzyme concentration vs time)
 - (1pts) Which linear line do you choose and why?
 - (1pts) What was the concentration of the enzyme you used for the remaining experiment?

- (2pts) **Figure 2b - Scatter plot graph (Michaelis-Menten) from kinetics of different substrate concentrations** (Velocity vs. substrate)
 - (1pts) What were your estimated $V_{\max}$ and K_m ?
 - (1pts) What was the concentration of the substrate you used for the remaining experiment?
- (2pts) **Figure 2c - Scatter plot graph (Lineweaver-Burke) from the inverse of figure 2b** (1/Velocity vs. 1/substrate)
 - (1pts) What were your calculated $V_{\max}$ and K_m ?
 - (1pts) Explain the relationship between $V_{\max}$ and K_m , and what do these values mean?
- **Figure 3. Determining the K_i of the inhibitor**
 - (2pts) **Figure 3a - Scatter plot graph (Michaelis-Menten) from kinetics of different inhibitor concentrations** (Velocity vs. inhibitor)
 - (1pts) What were your estimated inhibitor $V_{\max}$ and K_i ?
 - (1pts) What was the concentration of the inhibitor you used for the remaining experiment?
 - (2pts) **Figure 3b - Scatter plot graph (Michaelis-Menten) from kinetics of different inhibitor concentrations** (% inhibition vs. inhibitor)
 - (1pts) What were your estimated inhibitor $V_{\max}$ and K_i ?
 - (1pts) What was the concentration of the inhibitor you used for the remaining experiment?
 - (2pts) **Figure 3c - Scatter plot graph (Lineweaver-Burke) from the inverse of Figure 3a** (1/Velocity vs. 1/inhibitor)
 - (1pts) What were your calculated $V_{\max}$ and K_i ?
 - (1pts) Compare and contrast K_i to K_m , and what do these values mean?
- **Figure 4. Identifying the type of Inhibition**
 - (2pts) **Figure 4a - Scatter plot graph (Michaelis-Menten) from kinetics displaying inhibited vs noninhibited** (Velocity vs. inhibitor)
 - (2pts) Observe the relationship with the graphs, and explain the evidence showing whether your inhibitor is competitive vs non-competitive.
 - (2pts) **Figure 4b - Scatter plot graph (Lineweaver-Burke) from the inverse of Figure 4a** (1/Velocity vs. 1/inhibitor)
 - (2pts) Observe the relationship with the graphs, and explain the evidence showing whether your inhibitor is competitive vs non-competitive.
 - (2pts) **Figure 4c - Scatter plot graph (Relationship) from kinetics of displaying substrate vs inhibitor** (% inhibition vs. substrates)
 - (2pts) Observe the relationship with the graphs, and explain the evidence showing whether your inhibitor is competitive vs non-competitive.

Discussion (3 points)

- (2pts) What were the sources of error or limitations you ran into?
 - (1pts) What experiment could you improve?
-

Presentation Expectations

- At the end of the project, each group will present their poster to the class. Be prepared to:
 - **Explain your results and conclusions**
 - **Answer all questions**
 - The instructor can call on any student to answer his question.
 - Make sure that everyone in your team understands the project, not only their small part!

Grading Rubric – Total: 60 points

1. Results - Experimental Results – 22 points

- **All results are clearly presented** using graphs, tables, and labeled figures (Michaelis-Menten plots, Lineweaver-Burk plots, standard curves, kinetics, protein concentration).
- **Quantitative and/or statistical analysis** is present and appropriate.
- Visuals are properly labeled, include units, and support the group's conclusions.

2. Results - Q&A and Understanding of Concepts – 17 points

- Group demonstrates a **deep understanding** of their data and experimental methods.
- Students can **clearly explain** what their results mean and how they reached their conclusions.
- Confidently and accurately answer instructor questions

3. Background & Discussion – 8 points

- Objective and research question are clearly stated.
- Provides a **concise, accurate summary** of P450BM3, the chosen inhibitors (Imidazole, 4-Phenylimidazole), and substrate (Lauric Acid).
- Clearly explains the **significance** of determining inhibition type.

4. Methods (Flowchart Format) – 5 points

- Presents a clear, logical **flowchart** of the experimental steps.
- Includes major techniques: Bradford assay, Beer's Law, enzyme kinetics, protein purification, etc.
- Shows understanding of the **purpose** of each method in the overall workflow.

5. Presentation Quality & Visual Design – 8 points

- Poster is visually organized, clean, and easy to follow.
- Text is concise; fonts and colors enhance readability.
- Group delivers a **clear, engaging oral presentation**—each member contributes.
- Poster includes: **Title, Introduction, Methods, Results, Discussion, Conclusion, and References.**

Summary of Grading Criteria:

Category	Points
Experimental Results	22
Q&A and Scientific Reasoning	17
Background & Introduction	8
Methods (Flowchart)	5
Visual Design & Oral Presentation	8
Total	60

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10: Activity 3-1 - Michaelis-Menten Analysis of Substrate Concentration on Reaction Rate

Learning Objectives

- Define key terms related to enzyme kinetics, including substrate, velocity, $V_{\max}$, and K_M .
- Explain the relationship between substrate concentration and enzymatic reaction rate.
- Describe the purpose of the Michaelis-Menten equation.
- Apply Beer's Law to calculate enzyme concentration from spectrophotometric data.
- Recognize why optimizing enzyme and substrate concentrations is essential in kinetic studies.

Definition: Term

- **Enzyme:** A biological catalyst that speeds up chemical reactions without being consumed.
- **Substrate:** The specific molecule upon which an enzyme acts.
- **Product:** The molecule(s) formed from substrate transformation.
- **Reaction Rate (Velocity):** The speed at which the product is formed, typically measured as $\Delta[\text{product}]/\Delta\text{time}$.
- **$V_{\max}$:** The maximum rate of reaction when the enzyme is fully saturated with substrate.
- **K_M :** The substrate concentration at which the reaction rate is half of $V_{\max}$.
- **Michaelis-Menten Equation:** Describes how reaction velocity varies with substrate concentration.
- **Beer's Law:** A mathematical relationship used to determine concentration from absorbance: $A = \epsilon \times c \times l$.
- **Extinction Coefficient (ϵ):** A constant that indicates how strongly a substance absorbs light at a specific wavelength.
- **Cytochrome P450:** A family of heme-containing enzymes involved in metabolizing fatty acids, drugs, and toxins.

In-Lab Focus

- Use Beer's Law to calculate enzyme concentration.
- Vary both enzyme and substrate concentrations to observe their effects on reaction rate.
- Generate a Michaelis-Menten plot and estimate kinetic parameters ($V_{\max}$ and K_M).

Effect of Substrate Concentration on Reaction Rate

Background

In this lab series, you will analyze the enzymatic activity of a cytochrome P450 enzyme, a heme-containing protein involved in the metabolism of fatty acids, drugs, and other compounds. One of the most fundamental properties of any enzyme is its **kinetics**—how fast it catalyzes a reaction under different conditions. Understanding the enzyme's kinetics helps us characterize its efficiency, identify optimal reaction conditions, and compare its behavior under different physiological or experimental settings.

In this specific activity, you will investigate how **substrate concentration** affects the rate of the enzymatic reaction. This relationship is described by the **Michaelis-Menten equation**, which helps us determine two key kinetic parameters:

- **$V_{\max}$:** The maximum velocity the enzyme can achieve.
- **K_M :** The substrate concentration at which the reaction rate is half of $V_{\max}$.

Figuring out the best substrate concentration is critical. If the concentration is **too low**, the enzyme won't function at its full potential. If it's **too high**, you may saturate the enzyme, leading to wasted substrate and skewed data. By conducting this experiment and constructing a Michaelis-Menten plot, you will identify the optimal range of substrate concentrations and better understand how this enzyme behaves under physiological conditions.

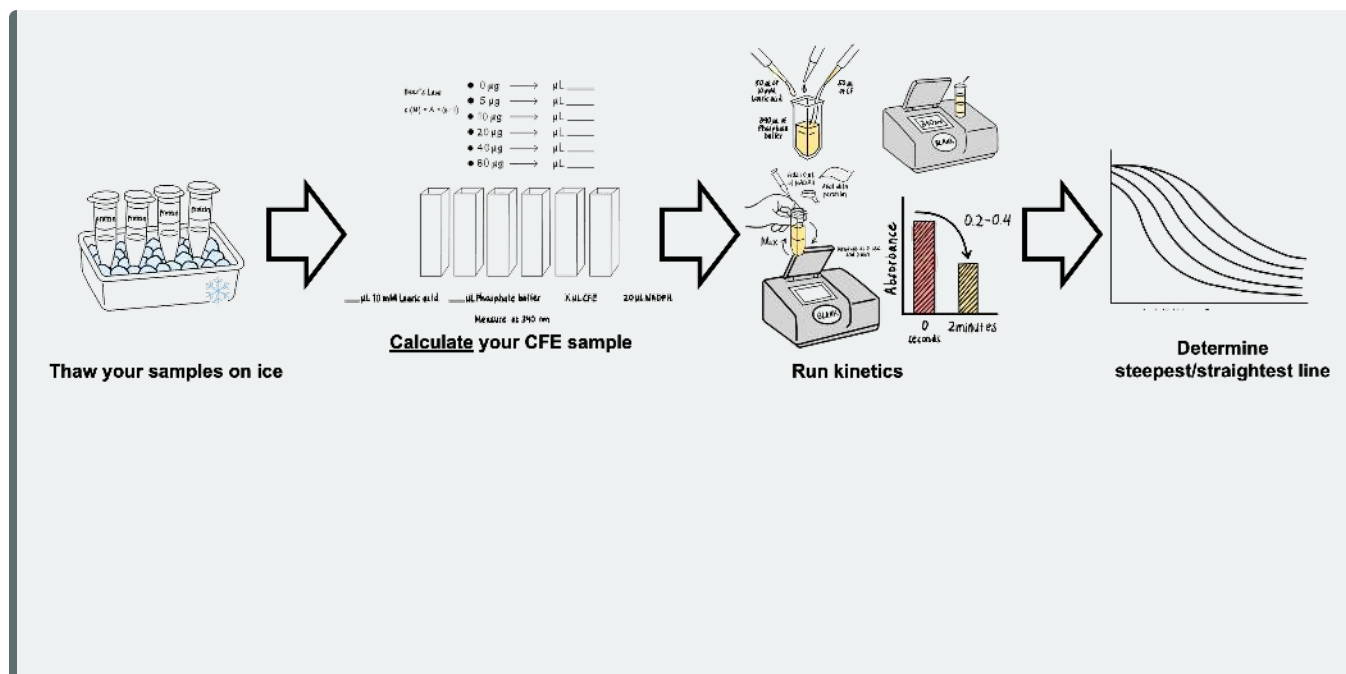
Before testing substrate concentration effects, you'll also learn how to calculate and optimize the amount of enzyme to use per reaction. You'll do this by applying **Beer's Law** to spectrophotometric data at 418 nm—an absorbance peak corresponding to the heme group in P450. This enables you to quantify the enzyme concentration accurately because each enzyme molecule contains one heme group.

Materials

- All materials used since the beginning of lab
- 100 mM potassium phosphate buffer, pH 7.4 (from 1 M stock) OR 0.5X PBS
- 10 mM lauric acid in 50 mM potassium carbonate (Carolina #871840)
- 10 mM NADPH (Fisher #481973100MG)
- Proteins (from previous experiments)
- UV-Visible Spectrophotometer (Thermo Scientific #840-300000)
- UVette cuvettes (Eppendorf #952010051)
- Micropipettes and sterile tips
- Parafilm

Procedures: Determine optimal Enzyme concentration for Kinetics Experiments

1. **Thaw your protein samples** on ice. Avoid repeated freeze-thaw cycles by preparing multiple aliquots.
2. **Use Beer's Law** to calculate the concentration of your enzyme (Follow Activity 1.3):
 - $[\text{Enzyme in uM}] = [(\text{Absorbance at 418 nm}) \div (91,000\text{M}^{-1}\text{cm}^{-1})] \times 10^{-6}$
 - Convert uM to mg/mL (same as ug/uL)
3. Based on this concentration, calculate the volume (in μL) needed to add the following amounts of enzyme to your cuvettes:
 - 0 ug/cuvette $\rightarrow$ _____ μL
 - 5 ug/cuvette $\rightarrow$ _____ μL
 - 10 ug/cuvette $\rightarrow$ _____ μL
 - 20 ug/cuvette $\rightarrow$ _____ μL
 - 40 ug/cuvette $\rightarrow$ _____ μL
 - 80 ug/cuvette $\rightarrow$ _____ μL
4. **Set up six (6) cuvettes** and follow the protocol from Activities 1-4. Make sure to adjust the volume of Enzymes and buffer for each cuvette tested.
 - 50 μL of 10 mM Lauric acid
 - _____ μL of Phosphate buffer
 - _____ μL of CFE
 - 20 μL of 10 mM NADPH (**Measure at 340nm**)
5. **Determine the optimal enzyme concentration**—Plot O.D. (absorbance) vs. time for each condition, and then look for the steepest AND linear increase over the 2-minute interval.



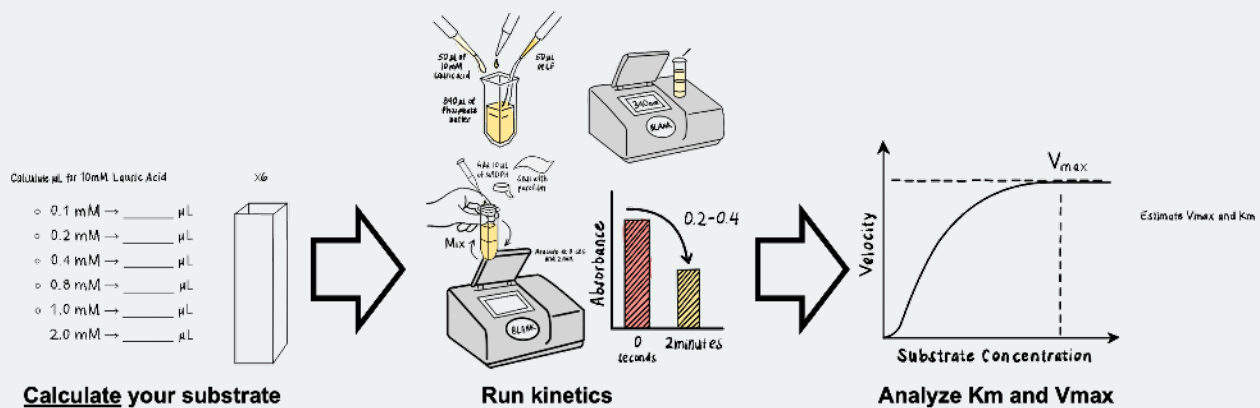


Image of a flow chart summarizing the Michaelis-Menten Analysis of Substrate Concentration on Reaction Rate. Image created by Diana Valdovinos.

Procedures: Determine K_m of substrates for Kinetics Experiments

Concerns: If you are low on Enzymes volumes, skip this experiment. Only go back here if you have any remaining enzyme after you determine your inhibitor type. Alternatively, you can decrease your enzyme volume by half or by a 3rd.

1. **Calculate** the volume (in μL) needed to add the following amounts of 10mM Lauric Acid (substrate) to your cuvettes:

- 0.05 mM (Optional) $\rightarrow$ μL
- 0.1 mM $\rightarrow$ μL
- 0.2 mM $\rightarrow$ μL
- 0.4 mM $\rightarrow$ μL
- 0.8 mM $\rightarrow$ μL
- 1.0 mM $\rightarrow$ μL
- 2.0 mM $\rightarrow$ μL

2. **Set up six (6) cuvettes** (or more if you want to do the additional optional steps).

- Follow the protocol from Activities 1-4.
 - You will use the **Optimal** enzyme concentration volume you found.
 - You will be adjusting the volume of Lauric Acid (substrate) and buffer for each cuvette tested.
 - Note: If you are low in enzymes, cut the enzyme volume in half.
 - μL of 10 mM Lauric acid
 - μL of Phosphate buffer
 - X μL of CFE
 - 20 μL of NADPH (Measure at 340nm)

3. **Determine the Velocity**—Plot Velocity vs. substrate concentration to create **Michaelis-Menten plots**

- Estimate **V_{max} and K_m** . The K_m concentration will be the one you use for the next experiment.

4. **(Optional)** Plot $1/\text{Velocity}$ vs. $1/\text{substrate concentration}$ to create **Lineweaver-Burke plots**

- (1pt) **V_{max} (1/b) =** _____
- (1pt) **K_m Absolute(1/b/m) =** _____

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11: Activity 3-2 - Determining the IC_{50} of Inhibitor

Learning Objectives

- Define and distinguish between **competitive** and **noncompetitive inhibition**.
- Explain the biochemical meaning of **IC_{50}** and **K_i** , and how they relate to inhibitor potency.
- Predict how changes in substrate or inhibitor concentrations affect enzyme activity.
- Understand the rationale behind choosing substrate and enzyme concentrations for inhibition assays.
- Describe the purpose and logic of a **2-fold serial dilution**.

Definition: Term

- **Enzyme Inhibition:** A process in which a molecule decreases enzyme activity by interfering with substrate binding or catalytic function.
- **Competitive Inhibitor:** A molecule that resembles the substrate and competes for binding at the enzyme's active site.
- **Noncompetitive Inhibitor:** A molecule that binds to a different site (not the active site) on the enzyme, affecting function regardless of substrate concentration.
- **IC_{50} :** The concentration of inhibitor that reduces enzyme activity by 50%.
- **K_i :** The inhibition constant; reflects the binding affinity of an inhibitor to the enzyme.
- **$P450_{BM3}$:** A bacterial cytochrome P450 enzyme used in this study for its monooxygenase activity.
- **V_{max} :** The maximum velocity of an enzymatic reaction at saturating substrate levels.
- **K_m :** The substrate concentration at which the reaction rate is half of V_{max} .
- **Serial Dilution:** A stepwise dilution where each step reduces the concentration by a fixed factor (e.g., 2-fold).

Pre-Lecture Tasks

- Review your prior enzyme kinetics data for $P450_{BM3}$ (K_m and V_{max} values).
- Watch a short video or animation on competitive vs. noncompetitive inhibition.
- Familiarize yourself with the spectrophotometer interface if you haven't used it before.
- Skim through the provided protocol and note any questions about the steps.

Measuring IC_{50} of Imidazole or 4-Phenylimidazole for $P450_{BM3}$

Background:

In previous experiments, you examined the kinetic properties of the $P450_{BM3}$ enzymes, focusing on the determination of Michaelis-Menten constants (K_m) and maximum reaction velocities (V_{max}). Today's experiment builds on that foundation by introducing the concept of enzyme **inhibition**—specifically, how small molecules can interfere with enzyme activity—and how to quantify this effect using **IC_{50}** and **K_i** values. The primary objective is to determine how inhibitors such as imidazole or 4-phenylimidazole affect the activity of the $P450_{BM3}$ enzyme, a bacterial cytochrome P450 known for its monooxygenase activity.

Enzyme inhibition occurs when a molecule (called an inhibitor) binds to an enzyme and decreases its catalytic activity. One common type is **competitive inhibition**, in which the inhibitor resembles the substrate and competes for binding at the enzyme's active site. This type of inhibition is reversible—meaning the inhibitor can bind and unbind—and is highly dependent on substrate concentration. When substrate levels are low, inhibitors can effectively block the active site and reduce enzyme activity. However, at high substrate concentrations, the substrate can "outcompete" the inhibitor for access to the active site, allowing the enzyme to function normally. Because of this, **V_{max} remains unchanged** in competitive inhibition, but **K_m increases**, indicating that a higher substrate concentration is needed to reach half the maximum velocity. This pattern can be visualized in a Lineweaver-Burk plot, where competitive inhibition shows a change in the x-intercept but not in the y-intercept.

In contrast, **noncompetitive inhibition** occurs when the inhibitor binds to a site other than the active site. This site can be present on either the free enzyme (E) or the enzyme-substrate complex (ES). Binding at this allosteric site forms an inactive enzyme-inhibitor complex (ESI) that cannot catalyze the reaction. Since the inhibitor does not compete with the substrate directly, increasing the substrate concentration does not reverse the inhibition. In noncompetitive inhibition, **V_{max} decreases** because some

enzymes are permanently taken out of the catalytic cycle by the inhibitor. However, K_m remains unchanged since the substrate's ability to bind is not directly affected.

In this activity, you will determine the IC_{50} —the concentration of inhibitor that reduces enzyme activity by 50%—for either imidazole or 4-phenylimidazole. These compounds are known inhibitors of P450 enzymes and act by binding to the oxygen-binding heme center, thereby interfering with enzymatic turnover. Based on the data, you will also calculate the K_i (inhibition constant), which reflects the binding affinity of the inhibitor for the enzyme and provides insight into the type and strength of inhibition. Understanding these parameters is crucial for interpreting how small molecules affect enzyme function and for designing selective inhibitors in research or pharmaceutical contexts.

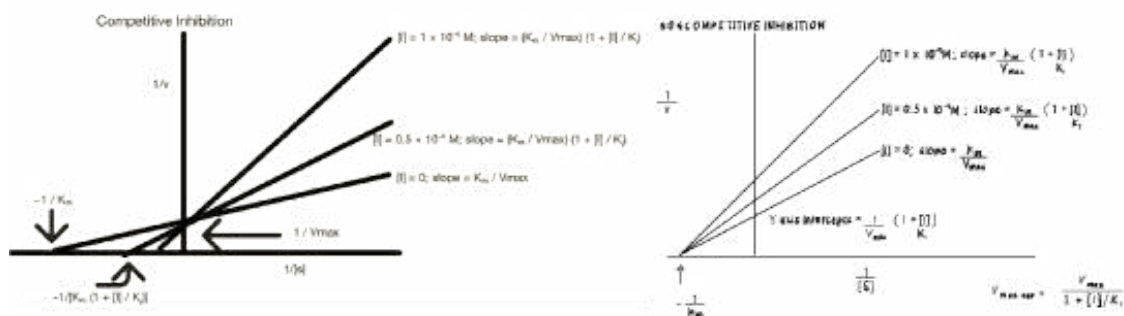
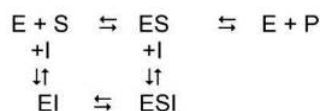


Image created by Diana Valdovinos.

Objective: To determine the inhibitory effect of imidazole and 4-phenylimidazole on the P450_{BM3} enzyme by calculating IC_{50} values using spectrophotometric assays.

Materials:

- 100 mM potassium phosphate buffer, pH 7.4 OR 0.5X PBS
- 10 mM lauric acid in 50 mM potassium carbonate (Carolina #871840)
- 10 mM NADPH (Fisher Scientific #481973100MG)
- 100 mM (or 100X) imidazole in 100 mM potassium phosphate OR 0.5X PBS (Fisher Scientific #A1022122)
- 100 μ M (or 100X) 4-phenylimidazole in 100 mM potassium phosphate OR 0.5X PBS (Fisher Scientific #L0223704)
- Purified P450_{BM3} protein (from previous experiments)
- UV/Visible Spectrophotometer (Thermo Scientific #840-300000)
- UVette Cuvettes (Eppendorf #952010051)
- Pipettes and pipette tips
- Incubator or water bath set to 37°C

Procedure: Determine K_i of inhibitors for Kinetics Experiments

Concerns: If you are low on Enzymes volumes, you can decrease your enzyme volume by half or by a 3rd.

1. **Calculate** the volume (in μ L) needed to add the following amounts of **100X inhibitor** to your cuvettes:

- 16X $\rightarrow$ _____ μ L (Optional)
- 8X $\rightarrow$ _____ μ L
- 4X $\rightarrow$ _____ μ L
- 2X $\rightarrow$ _____ μ L
- 1X $\rightarrow$ _____ μ L
- 0.5X $\rightarrow$ _____ μ L
- 0.25X $\rightarrow$ _____ μ L
- 0.125X $\rightarrow$ _____ μ L (Optional)
- 0.06125X $\rightarrow$ _____ μ L (Optional)

Note: You need to know the total volume of your reaction. Apply $C_1V_1 = C_2V_2$. For example, if your total volume is 500 μL (from the standard protocol), then 16X would be $100X/16X = 6.25\text{-fold}$ dilution. $500\mu\text{L} \div 6.25\text{-fold} = \underline{80\ \mu\text{L}}$ of inhibitor to add to your reaction.

1. **Set up six (6) cuvettes** (or more if you want to do the additional optional steps).
2. **Follow the protocol from Activities 1-4. However, you will add Enzymes last, instead of NADPH**
 - You will use the **Optimal** enzyme concentration volume and **K_m** substrate volume you found in Activities 3.1.
 - You will be adjusting the volume of Inhibitors and buffer for each cuvette tested.
 - X μL of 10 mM Lauric acid
 - X μL of CFE
 - 20 μL of NADPH (Measure at 340nm)
 - ____ μL of Phosphate buffer
 - ____ μL of inhibitor
3. **Determine the Velocity**—Plot Velocity vs. inhibitor concentration to create Michaelis-Menten plots
 - Estimate V_{max} and K_i. The K_i concentration will be the one you use for the next experiment.

Learning Objectives

After completing this experiment and analysis, you should be able to:

- Perform a serial dilution and correctly set up a dose-response inhibition assay.
- Accurately measure changes in enzyme activity using spectrophotometric readings.
- Calculate and interpret IC₅₀ values from inhibition curves.
- Differentiate between competitive and noncompetitive inhibition based on kinetic patterns (V_{max} and K_m changes).
- Use % inhibition data to draw conclusions about inhibitor potency and mechanism.
- Relate your findings to broader applications, such as drug design or enzyme regulation in cells.

Data Summary Table (To Complete After Lab)

Cuvette #	Inhibitor Concentration (μM)	Absorbance Slope	% Inhibition
1			
2			
...			
9			

11: Activity 3-2 - Determining the IC₅₀ of Inhibitor is shared under a [not declared](#) license and was authored, remixed, and/or curated by LibreTexts.

12: Activity 3-3 - Identifying the Type of Enzyme Inhibition

Learning Objectives

- Define and distinguish between **competitive** and **noncompetitive** enzyme inhibition.
- Interpret the effect of substrate concentration on percent inhibition.
- Analyze experimental data to determine the **type of inhibition**.
- Use graph-based results to calculate the **inhibition constant (K_i)**.

Definition: Term

- **Enzyme Inhibitor:** A molecule that binds to an enzyme and decreases its activity.
- **Competitive Inhibition:** Inhibitor competes with the substrate for the enzyme's active site. It increases K_m but does not affect V_{max} .
- **Noncompetitive Inhibition:** Inhibitor binds to an allosteric site (not the active site), decreasing V_{max} but usually not affecting K_m .
- **IC₅₀:** The inhibitor concentration that causes 50% inhibition of enzyme activity.
- **K_i (Inhibition Constant):** A measure of inhibitor binding affinity. Lower K_i = stronger inhibitor.
- **% Inhibition:** A calculated value that shows how much the inhibitor reduces enzyme activity, based on slope comparisons.

Note

- Review your IC₅₀ data from previous experiments.
- Understand how a **2-fold serial dilution** works and why it's used to generate a range of substrate concentrations.
- Be prepared to **work quickly** when adding enzyme to each cuvette to maintain consistent reaction timing.

Identifying the Type of Enzyme Inhibition (Competitive vs. Noncompetitive)

In this experiment, you will determine whether an inhibitor behaves as a **competitive** or **noncompetitive** inhibitor by analyzing how the **percent inhibition** changes with varying substrate concentrations. This activity builds on your previous work in which you estimated the IC₅₀, the inhibitor concentration that causes 50% inhibition of enzyme activity. To understand the difference between **competitive** and **noncompetitive** inhibition, recall the following:

- **Competitive Inhibition:** The inhibitor competes with the substrate for the enzyme's active site. At low substrate concentrations, the inhibitor significantly reduces enzyme activity. However, as substrate concentration increases, the substrate can outcompete the inhibitor, leading to a **decrease in percent inhibition**. On a graph of **% inhibition vs. substrate concentration**, this appears as a **downward-sloping line**. The maximum reaction velocity (V_{max}) remains **unchanged**, but the apparent affinity for the substrate (K_m) increases.
- **Noncompetitive Inhibition:** The inhibitor binds to a site other than the active site, and it can bind whether or not the substrate is bound. This results in a **constant percent inhibition**, regardless of substrate concentration, because increasing the substrate does not displace the inhibitor. The graph shows a **horizontal line**, indicating that the inhibitor affects V_{max} (lowers it) and usually **does not affect K_m** .

In this experiment, you will graph **% inhibition vs. substrate concentration** using your experimental data. Based on the shape of the curve:

- A **flat line** suggests **noncompetitive inhibition**.
- A **declining line** suggests **competitive inhibition**.

By comparing your curve to these expected patterns, you can **identify the inhibition type** and then calculate the **inhibition constant (K_i)**. If you do not have access to your previous lab data, you may substitute in data from your group or another team.

Materials

- 100 mM potassium phosphate buffer, pH 7.4 OR 0.5X PBS
- 10 mM lauric acid in 50 mM potassium carbonate (Carolina #871840)

- 10 mM NADPH (Fisher Scientific #481973100MG)
- 100 mM imidazole in 100 mM potassium phosphate OR 0.5X PBS (Fisher Scientific #A1022122)
- 100 μ M 4-phenylimidazole in 100 mM potassium phosphate OR 0.5X PBS (Fisher Scientific #L0223704)
- Purified P450BM3 protein (from previous experiments)
- UV/Visible Spectrophotometer (Thermo Scientific #840-300000)
- UVette Cuvettes (Eppendorf #952010051)
- Pipettes and pipette tips
- Parafilm

Procedure: Determine the type of inhibitors for Kinetics Experiments

Concerns: If you are low on Enzymes volumes, you can decrease your enzyme volume by half or by a 3rd.

1. **Calculate the volume (in μ L)** needed to add the following amounts of Lauric Acid (Substrates) to your cuvettes:
 - x4 $\rightarrow$ _____ μ L
 - x2 $\rightarrow$ _____ μ L
 - x1 $\rightarrow$ (This is the volume of your K_m)
 - $\div 2 \rightarrow$ _____ μ L
 - $\div 4 \rightarrow$ _____ μ L
2. **Set up ten (10) cuvettes.**
3. **Without inhibitor:** Follow the protocol from Activities 1-4. However, you will **add Enzymes last**, instead of NADPH
 - You will use the **Optimal** enzyme concentration volume from Activities 3.1.
 - You will be adjusting the volume of substrates and buffer for each cuvette tested.
4. **With Inhibitors:** Follow the same protocol from Step 3.
 - You will use the **K_i** inhibitor volume from Activities 3.2.
 - You will be adjusting the volume of substrates and buffer for each cuvette tested.
5. **Determine the Velocity**
6. Determine the inhibition percentages at each substrate concentration.
 - Percent inhibition = $100 \times (V_{\text{noninhibited}} - V_{\text{inhibited}}) / V_{\text{noninhibited}}$
7. Plot Percent inhibition vs. Substrate concentration to a line graph.
 - A **flat line** suggests **noncompetitive inhibition**.
 - A **downward slope** suggests **competitive inhibition**.

Cuvette #	10mM NADPH (uL)	Lauric Acid substrates (uL)	Enzyme CFE (uL)	• Imidazole (mM) • 4-phenylimidazole (uM)	PO ₄ buffer (uL)	Total Volume (uL)
1	100	x4	previous result	IC ₅₀		500
2	100	x2	previous result	IC ₅₀		500
3	100	x1 (previous result)	previous result	IC ₅₀		500
4	100	$\div 2$	previous result	IC ₅₀		500
5	100	$\div 4$	previous result	IC ₅₀		500

Interpreting the Sample Graph

In the sample graph shown, two sets of data are plotted:

- The **X symbols** represent competitive inhibition. The line slopes downward as substrate concentration increases, indicating that more substrate overcomes inhibition.
- The **O symbols** represent noncompetitive inhibition. The line is flat, showing that substrate concentration does not affect the degree of inhibition.

This graph provides a visual key for interpreting your own results.

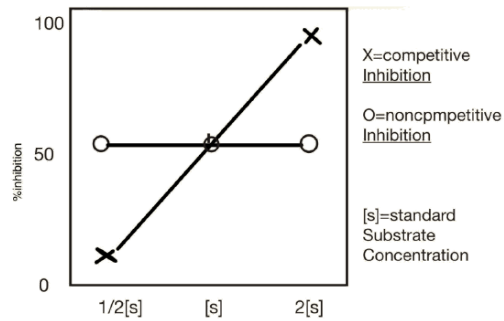


Image created by Diana Valdovinos.

12: Activity 3-3 - Identifying the Type of Enzyme Inhibition is shared under a [not declared](#) license and was authored, remixed, and/or curated by LibreTexts.

Materials

The materials list below assumes there will be 6 groups (~24 students).

Required Cell lines

- Quantity of 1 – Frozen stocks of BL21 E. coli containing T7BM3 plasmid
- Quantity of 1 – Frozen stocks of BL21 E. coli containing pET3a plasmid

Required Equipment and Materials (or Equivalence)

- Quantity of 6 - Micropipette sets - P20, P200, P1000 micropipettes
- Quantity of 6 – Full Boxes of p1000 tips and p200 tips
- Quantity of 6 - Serological pipettors (25mL) – (Carolina, #736871)
- Quantity of 1 –24/50 counts is required - 96-well plates per student (Carolina, #216732)
- Quantity of 3 - Centrifuge machine with rotor (Carolina, # 217221 with #217234)
- Quantity of 6 - Analytical scales (Carolina, #702547)
- Quantity of 6 - Disinfect spray or Ethanol spray (Flinn Scientific, #E0013)
- Quantity of 1 - Large Biohazard waste container
- Quantity of 1 - Hazardous liquid waste container
- Quantity of 6 - Microcentrifuge machine (Carolina, # 214061)
- Quantity of 12 - Centrifuge holder racks
- Quantity of 1 - Shaking incubator (Carolina, #214745)
- Quantity of 1 – Liquid Nitrogen Freezer OR -80°C Freezer
- Quantity of 6 - Vortex
- Quantity of 6 - Ice buckets or Styrofoam containers
- Quantity of 6 – Spectrophotometry
- Quantity of 1 – Scissors
- Quantity of 12 - 50mL beakers
- Quantity of 6 - 1L beakers
- Quantity of 6 - Six small stir bars (Carolina, # 701091)
- Quantity of 1 - Stir bars Retriever (Carolina, # 701089)
- Quantity of 12 - Dialysis clips (Carolina, #684239)
- Quantity of 12 - Filtration column chromatography with stopcock
- Quantity of 3 – SDS-PAGE Electrophoresis (Bio-Rad, #1658005)
- Quantity of 3 – SDS-PAGE Power supply (Bio-Rad, # 1645050)
- Quantity of 1 - Gel opening lever (Fisher Scientific, #NC0779477)
- Quantity of 1 - ChemiDoc MP Imaging System (Bio-Rad, # 12003154)
- Quantity of 12 – Fine-point Sharpies
- Quantity of 5 - Different color Expo markers (This is for the instructor)

Materials that only need to be purchased when inventory is low (or equivalent)

- Quantity of 1 – 100/500 counts is required - 50mL **High Speed** conical tubes (Fisher Scientific, 352070).
- Quantity of 1 – 20/100 grams is required - Lauric acid (Carolina, #871840)

- Quantity of 1 – 100/500 grams is required - Potassium carbonate (Carolina, #882836)
- Quantity of 1 – 50/100 counts is required – Weight boats (Carolina, # 215594)
- Quantity of 1 – 5/10 grams is required - Sephadex-G25 (Fisher Scientific, #50-177-8947)
- Quantity of 1 – 2/5 liter is required - 10X running buffer (Fisher Scientific #NC9684733)
- Quantity of 1 – 250/500g is required - Luria-Bertani Broth solid (Carolina, #216710)
- Quantity of 1 – 5/10 count is required - 10% TGX stain-free gels (Fisher Scientific, #NC1085832)
- Quantity of 1 – 2/10mL is required - 4X sample buffer (Fisher Scientific, # NC0596823)
- Quantity of 1 – 0.1/1mL is required - Protein Ladder (Fisher Scientific, #50-901-2665)

Required Materials that must be purchased before the class starts (or equivalent)

- Quantity of 1 – 150 counts - Sterile 25mL Serological pipette tips (Carolina, #214434)
- Quantity of 1 – 500 counts – Sterile 1.5mL Microcentrifuge tubes (Carolina, #215244)
- Quantity of 1 – 100 counts - Sterile cuvettes (Carolina, #215087)
- Quantity of 1 – 10 ft – Parafilm (Carolina, # 513641)
- Quantity of 1 - 500g - Ammonium sulfate (Carolina, #844362)
- Quantity of 1 – 10 ft - Dialysis tubing (Carolina, #684212)
- Quantity of 1 – 100mL - 100X ampicillin (Carolina, #216860)
- Quantity of 1 – 250 prep - Bacterial cell lysis kit (GoldBio.com, #GB-176-250)
- Quantity of 1 – 500mL kit - Pierce BCA Protein Assay Kit (Fisher Scientific, # PI23227)
- Quantity of 1 – 1L – 10X PBS, pH 7.4
- Quantity of 1 - 250mg - NADPH (Fisher Scientific #48-197-3100MG)

Index

D

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Overview

Title: Lab manual : Protein Biomanufacturing and Biochemistry (Biot274 @IVC)

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- Lab manual : Protein Biomanufacturing and Biochemistry (Biot274 @IVC) - *Undeclared*
 - Front Matter - *Undeclared*
 - TitlePage - *Undeclared*
 - InfoPage - *Undeclared*
 - Table of Contents - *Undeclared*
 - Licensing - *Undeclared*
 - 1: Activity 1-1 - Protein Extraction Using Bacterial Cell Lysis - *Undeclared*
 - 2: Activity 1-2 - Bradford Assay – Measuring Protein Concentration - *Undeclared*
 - 3: Activity 1-3 - Beer's Law Spectrophotometric Analysis - *Undeclared*
 - 4: Activity 1-4 - Analysis of Enzymatic Activities of Cell-Free Extracts - *Undeclared*
 - 5: Activity 2-1 - Purification via Ammonium Sulfate Saturation of Cell-Free Extract - *Undeclared*
 - 6: Activity 2-2 - Dialysis for Enzyme Purification - *Undeclared*
 - 7: Activity 2-3 - Size Exclusion Gel Filtration Purification of Enzymes - *Undeclared*
 - 8: Activity 2-4 - Verification of Protein Purification via Polyacrylamide Gel Electrophoresis - *Undeclared*
 - 9: Activity 3-0 - Project Instructions - Investigating Enzyme Kinetics - *Undeclared*
 - 10: Activity 3-1 - Michaelis-Menten Analysis of Substrate Concentration on Reaction Rate - *Undeclared*
 - 11: Activity 3-2 - Determining the IC₅₀ of Inhibitor - *Undeclared*
 - 12: Activity 3-3 - Identifying the Type of Enzyme Inhibition - *Undeclared*
 - Materials - *Undeclared*
 - Index - *Undeclared*
 - Glossary - *Undeclared*
 - Detailed Licensing - *Undeclared*
 - Detailed Licensing - *Undeclared*

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 - TitlePage - *Undeclared*
 - InfoPage - *Undeclared*
 - Table of Contents - *Undeclared*
 - Licensing - *Undeclared*
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 - 2: Activity 1-2 - Bradford Assay – Measuring Protein Concentration - *Undeclared*
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 - 4: Activity 1-4 - Analysis of Enzymatic Activities of Cell-Free Extracts - *Undeclared*
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 - 6: Activity 2-2 - Dialysis for Enzyme Purification - *Undeclared*
 - 7: Activity 2-3 - Size Exclusion Gel Filtration Purification of Enzymes - *Undeclared*
 - 8: Activity 2-4 - Verification of Protein Purification via Polyacrylamide Gel Electrophoresis - *Undeclared*
 - 9: Activity 3-0 - Project Instructions - Investigating Enzyme Kinetics - *Undeclared*
 - 10: Activity 3-1 - Michaelis-Menten Analysis of Substrate Concentration on Reaction Rate - *Undeclared*
 - 11: Activity 3-2 - Determining the IC₅₀ of Inhibitor - *Undeclared*
 - 12: Activity 3-3 - Identifying the Type of Enzyme Inhibition - *Undeclared*
 - Materials - *Undeclared*
 - Index - *Undeclared*
 - Glossary - *Undeclared*
 - Detailed Licensing - *Undeclared*
 - Detailed Licensing - *Undeclared*

Index

D

dire

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Overview

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 - Front Matter - *Undeclared*
 - TitlePage - *Undeclared*
 - InfoPage - *Undeclared*
 - Table of Contents - *Undeclared*
 - Licensing - *Undeclared*
 - 1: Activity 1-1 - Protein Extraction Using Bacterial Cell Lysis - *Undeclared*
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 - 3: Activity 1-3 - Beer's Law Spectrophotometric Analysis - *Undeclared*
 - 4: Activity 1-4 - Analysis of Enzymatic Activities of Cell-Free Extracts - *Undeclared*
 - 5: Activity 2-1 - Purification via Ammonium Sulfate Saturation of Cell-Free Extract - *Undeclared*
 - 6: Activity 2-2 - Dialysis for Enzyme Purification - *Undeclared*
 - 7: Activity 2-3 - Size Exclusion Gel Filtration Purification of Enzymes - *Undeclared*
 - 8: Activity 2-4 - Verification of Protein Purification via Polyacrylamide Gel Electrophoresis - *Undeclared*
 - 9: Activity 3-0 - Project Instructions - Investigating Enzyme Kinetics - *Undeclared*
 - 10: Activity 3-1 - Michaelis-Menten Analysis of Substrate Concentration on Reaction Rate - *Undeclared*
 - 11: Activity 3-2 - Determining the IC₅₀ of Inhibitor - *Undeclared*
 - 12: Activity 3-3 - Identifying the Type of Enzyme Inhibition - *Undeclared*
 - Materials - *Undeclared*
 - Index - *Undeclared*
 - Glossary - *Undeclared*
 - Detailed Licensing - *Undeclared*
 - Detailed Licensing - *Undeclared*

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Title: Lab manual : Protein Biomanufacturing and Biochemistry (Biot274 @IVC)

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 - Front Matter - *Undeclared*
 - TitlePage - *Undeclared*
 - InfoPage - *Undeclared*
 - Table of Contents - *Undeclared*
 - Licensing - *Undeclared*
 - 1: Activity 1-1 - Protein Extraction Using Bacterial Cell Lysis - *Undeclared*
 - 2: Activity 1-2 - Bradford Assay – Measuring Protein Concentration - *Undeclared*
 - 3: Activity 1-3 - Beer's Law Spectrophotometric Analysis - *Undeclared*
 - 4: Activity 1-4 - Analysis of Enzymatic Activities of Cell-Free Extracts - *Undeclared*
 - 5: Activity 2-1 - Purification via Ammonium Sulfate Saturation of Cell-Free Extract - *Undeclared*
 - 6: Activity 2-2 - Dialysis for Enzyme Purification - *Undeclared*
 - 7: Activity 2-3 - Size Exclusion Gel Filtration Purification of Enzymes - *Undeclared*
 - 8: Activity 2-4 - Verification of Protein Purification via Polyacrylamide Gel Electrophoresis - *Undeclared*
 - 9: Activity 3-0 - Project Instructions - Investigating Enzyme Kinetics - *Undeclared*
 - 10: Activity 3-1 - Michaelis-Menten Analysis of Substrate Concentration on Reaction Rate - *Undeclared*
 - 11: Activity 3-2 - Determining the IC₅₀ of Inhibitor - *Undeclared*
 - 12: Activity 3-3 - Identifying the Type of Enzyme Inhibition - *Undeclared*
 - Materials - *Undeclared*
 - Index - *Undeclared*
 - Glossary - *Undeclared*
 - Detailed Licensing - *Undeclared*
 - Detailed Licensing - *Undeclared*